# Why Can't You Always Trade When You Want To?

A visual guide to the hidden costs of buying and selling

## What if selling something was harder than you thought?

Imagine you're selling lemonade on a hot summer day. Customers line up, you sell a cup every few seconds, and the price stays steady. That's a liquid market.

Now try selling a limited-edition Nintendo Switch. Few buyers, all lowballing. To sell quickly, you'd have to drop the price. That's illiquid.

🍋

### The Lemonade Stand

Hot day, busy street. Buyers everywhere. Steady price, instant sales, any quantity.

HIGH LIQUIDITY

🎮

### The Rare Console

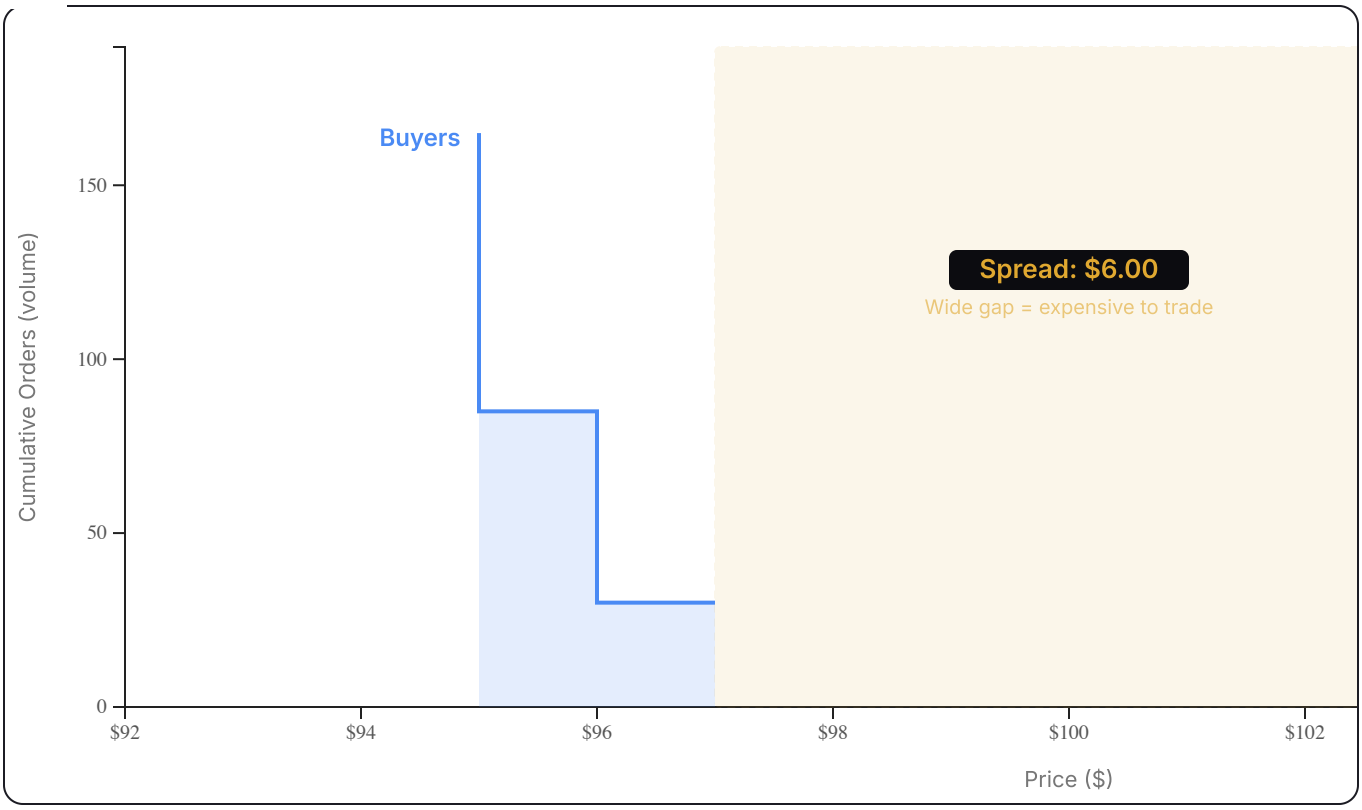
Few interested buyers. Each wants a discount. Selling fast = accepting a worse price.

LOW LIQUIDITY

Financial markets work the same way. When you want to buy or sell, there's always a gap between what buyers are willing to pay and what sellers are asking for. That gap is called the spread, and it's the hidden cost of every trade — a liquidity tax you pay every time you buy or sell.

The chart below is a depth chart — the standard way to visualize an order book. The blue area shows cumulative buy orders (higher at lower prices — buyers want to pay less). The red area shows cumulative sell orders (higher at higher prices — sellers want to charge more). The yellow gap between them is the spread. Toggle between the two states and watch what happens.

Liquid Market | Illiquid Market




**WHY A DEPTH CHART?** This is the canonical order book visualization used on every exchange. X-axis = price, Y-axis = cumulative volume available at that price or better. The gap between the two curves is the spread — the cost of trading. The D3.js animated transition (Munzner, Ch. 13) lets you *see* liquidity vanishing: the curves flatten, spread apart, and the gap widens. Both axes encode meaningful quantities — nothing is decorative.

We're going to explore five questions that unpack why you can't always trade when you want to — using real market data from stocks, currencies, commodities, and crypto.

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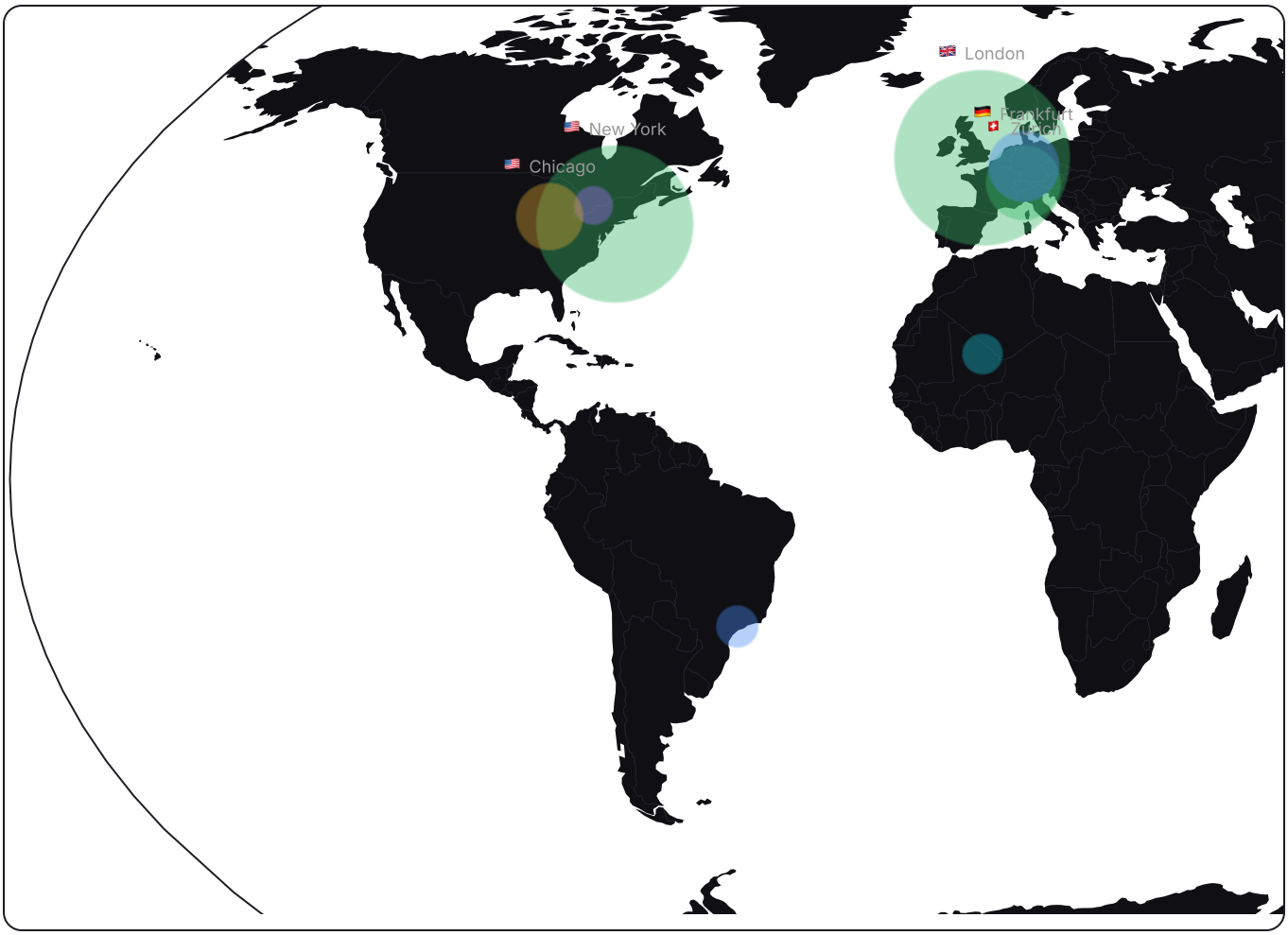
## Where does trading happen?

Before we look at *when* markets are liquid, it helps to understand *where* they are and *how big* they are. Financial markets are spread across the globe — and they vary enormously in scale.

Each circle below represents a major financial center. The size shows daily trading volume. Pick an asset class to see where it's traded — and how the map changes dramatically.

All | Foreign Exchange | Equities | Bonds | Commodities | Crypto




| $7.5T | $900B | $500B | $130B |
|---|---|---|---|
| FX DAILY VOLUME | US BONDS DAILY | US EQUITIES DAILY | GOLD DAILY |

$30B
BITCOIN DAILY

> *The foreign exchange market is larger than everything else combined. This matters for liquidity: FX pairs like EUR/USD are so deeply traded that spreads are measured in fractions of a cent. Smaller markets — agricultural commodities, crypto — have far less volume, which means wider spreads and higher trading costs.*

**WHY A MAP?** The next section asks "does every market have the same daily rhythm?" The answer is rooted in geography: markets follow the sun. Tokyo opens first, then London, then New York. This map sets the spatial context that makes the clock chart (next) immediately readable. Circle area is proportional to volume (Tufte's principle of proportional ink).

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## Q1: Does every market have the same daily rhythm?

Markets aren't equally active throughout the day. Most people assume trading is constant — but every market has quiet hours and busy hours. Let's look at when trading actually happens.

### 1.1 — Do stocks, currencies, and commodities all get busy at the same times?

Pick any ticker below. US equities show the classic U-shape — heavy at open, quiet at lunch, heavy before close. But switch to EUR/USD or Gold and the shape is completely different.

US EQUITIES (ET)

SPY AAPL TSLA MSFT GME

CURRENCIES (UTC)

EUR/USD GBP/USD AUD/USD

INTERNATIONAL INDICES (UTC)

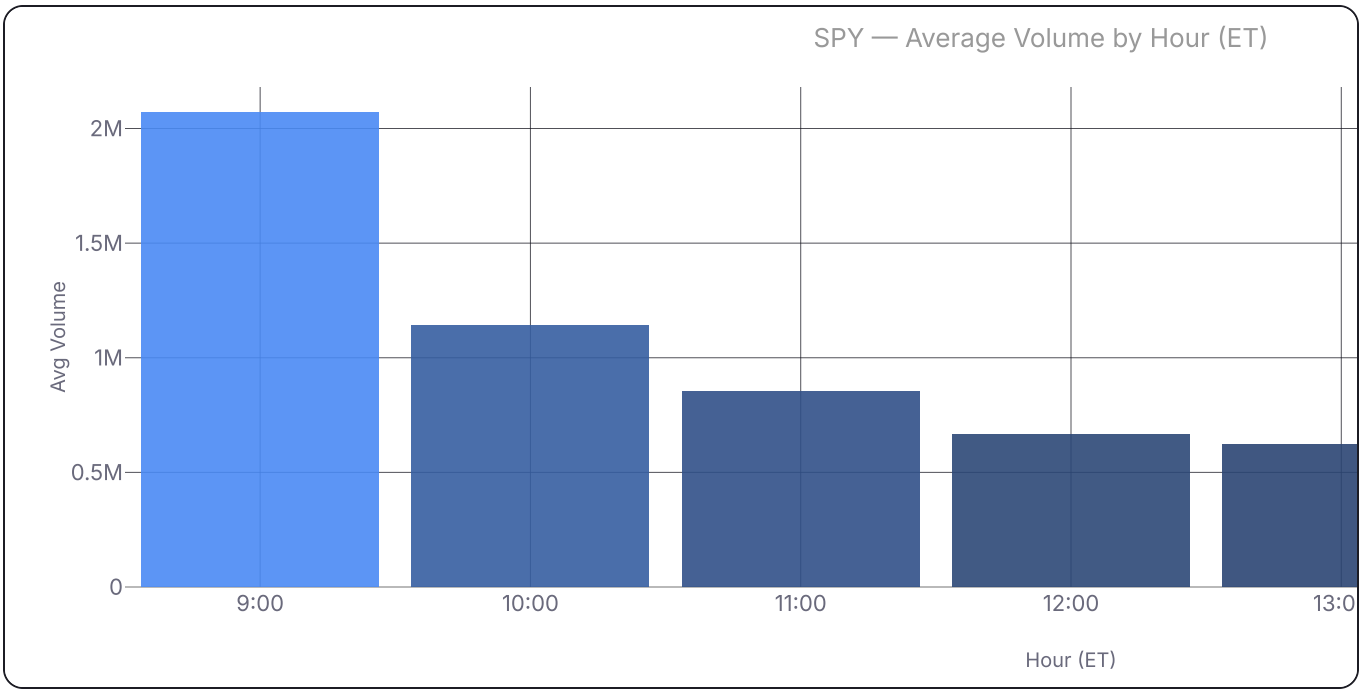
S&P 500 | Nikkei 225 | FTSE 100 | CAC 40 | ASX 200

BONDS (UTC)

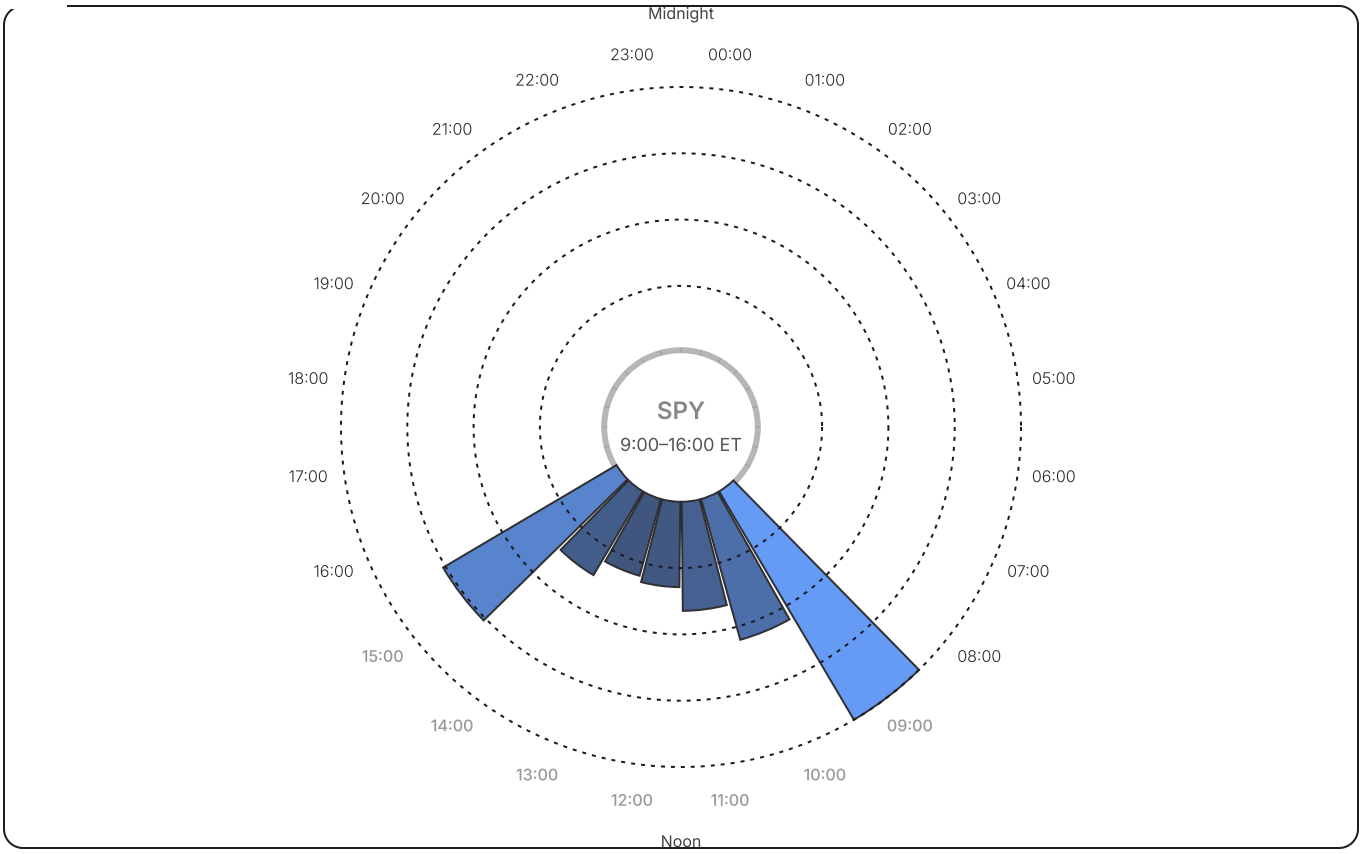
US 10Y Bond | German Bund | UK Gilt

COMMODITIES (UTC)

Gold | Oil | Nat Gas | Wheat | Soybeans | Sugar



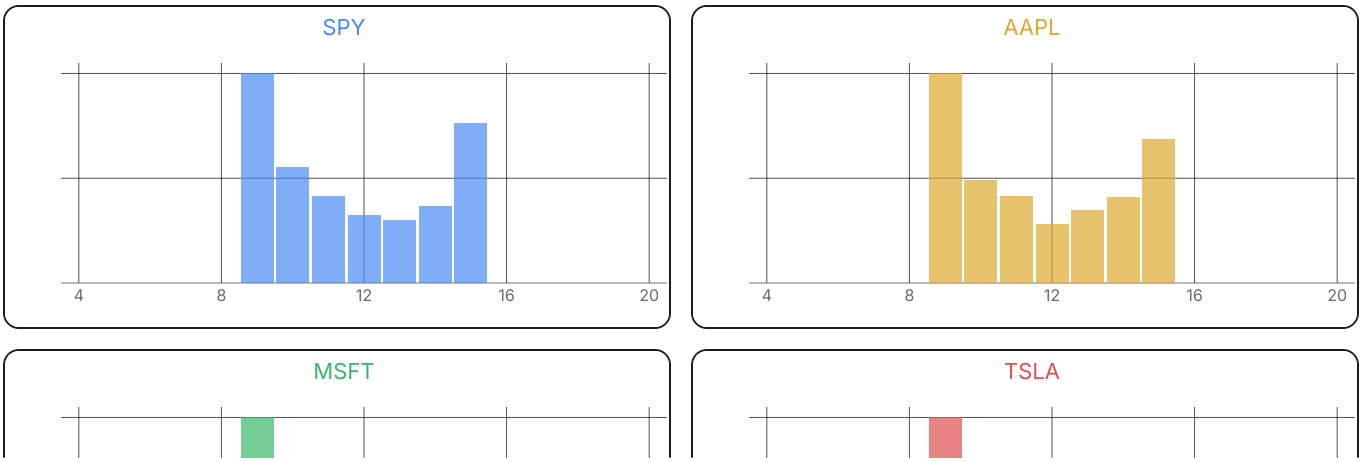

Now see the same data as a clock. Each hour is a spoke — the further it extends, the more trading volume at that hour. Color intensity encodes volume too. Switch assets and watch the clock reshape.

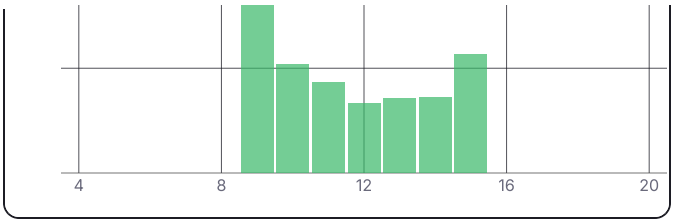


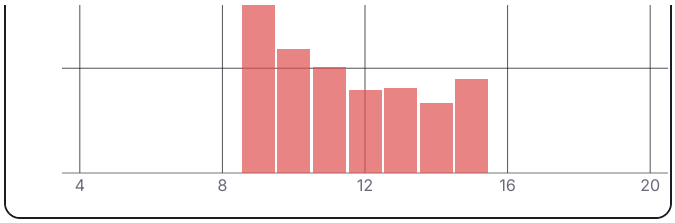

**WHY A RADIAL CHART?** Time is cyclical — 23:00 is next to 00:00, not at the far end of a bar chart. A radial layout respects this topology (Munzner, Ch. 7: spatial channel mappings). It also creates a memorable shape: the "fat" hours and "thin" hours form a silhouette that's instantly recognizable per asset class. The dark sectors for equities visually encode "this market is closed" — something a bar chart can't show.

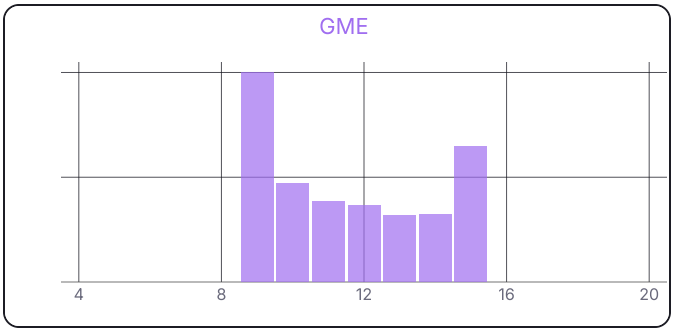
## 1.2 — Are there markets that never sleep vs ones with clear "office hours"?

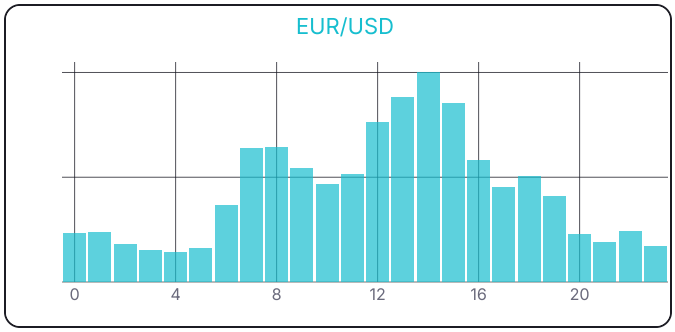
The small multiples below show every market's rhythm side by side, normalized to each one's peak. Compare the shapes: equities have a sharp U-shape within a narrow window, while EUR/USD and Gold trade around the clock with a broad hump during London/New York hours.

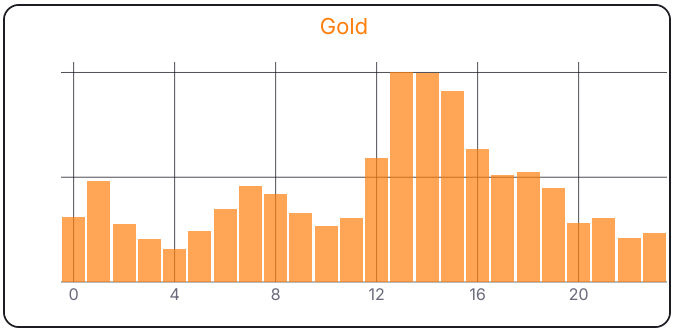


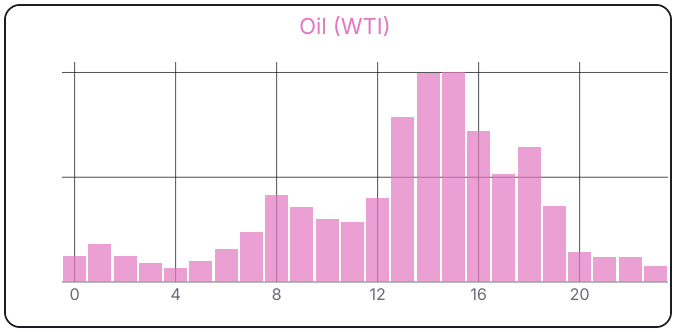


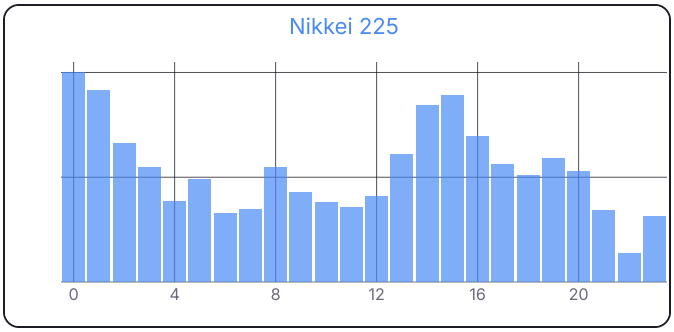



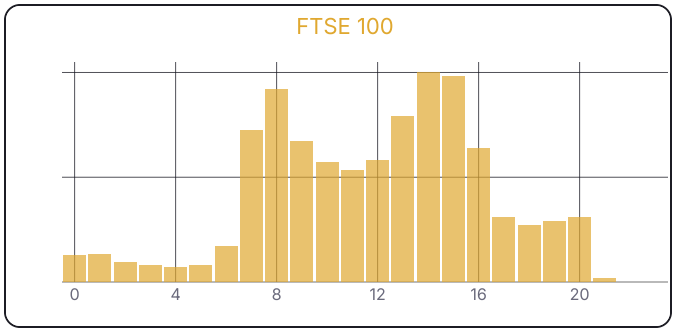



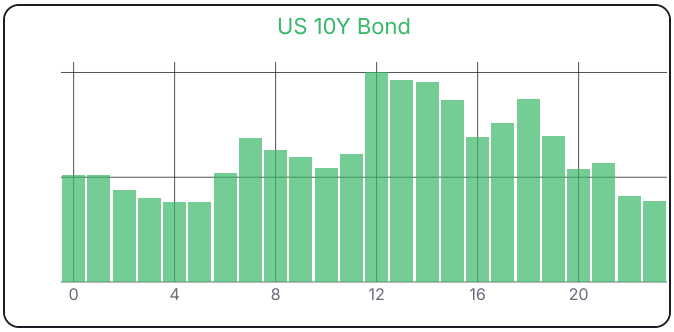



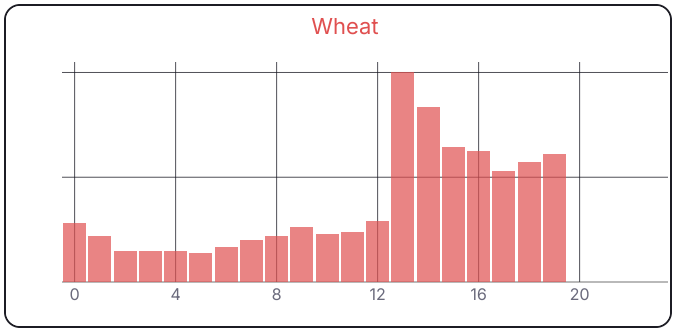



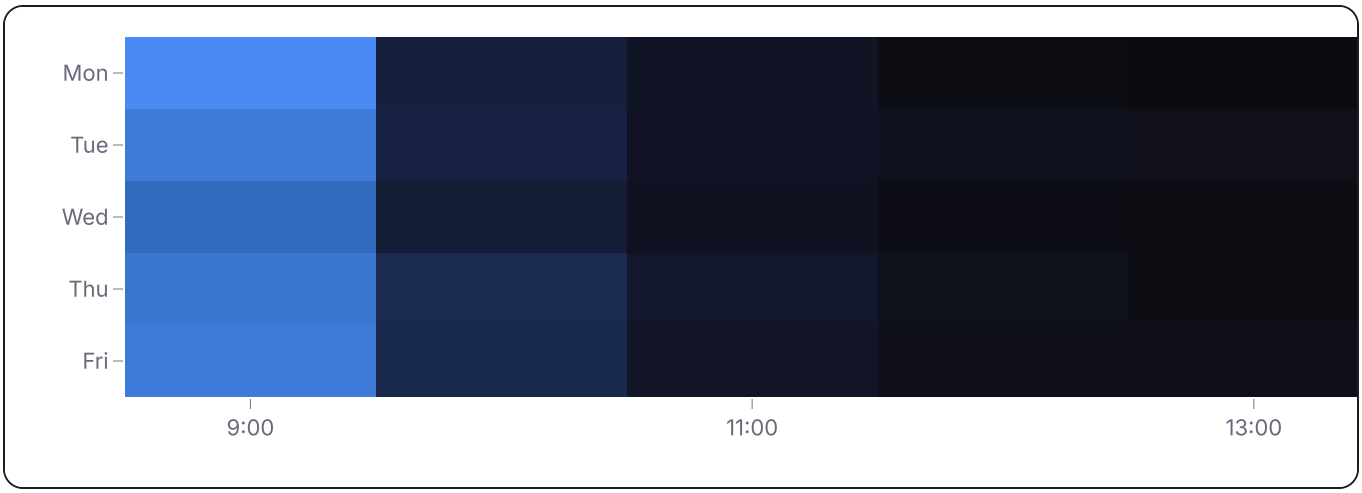




**WHY NORMALIZE?** Absolute volume varies by orders of magnitude across assets (millions of shares vs dozens of FX ticks). Normalizing to each ticker's peak lets us compare the *shape* of the rhythm — following Tufte's "small multiples" principle (*The Visual Display of Quantitative Information*, 1983): same structure, different data, enabling comparison of form rather than magnitude.

### 1.3 — Does the day of the week matter — is Monday different from Friday?

The heatmap encodes two dimensions at once: hour of day and day of week. Darker = more volume. Look for patterns: Are Mondays busier than Fridays? Are there dead zones where almost nothing trades?




> ***Answer:*** *Yes — Monday opens tend to be volatile (weekend news builds up), while Friday afternoons thin out as traders close positions for the weekend. Agricultural commodities like Sugar and Wheat have the narrowest trading windows — miss their peak hours and you're trading in a desert.*

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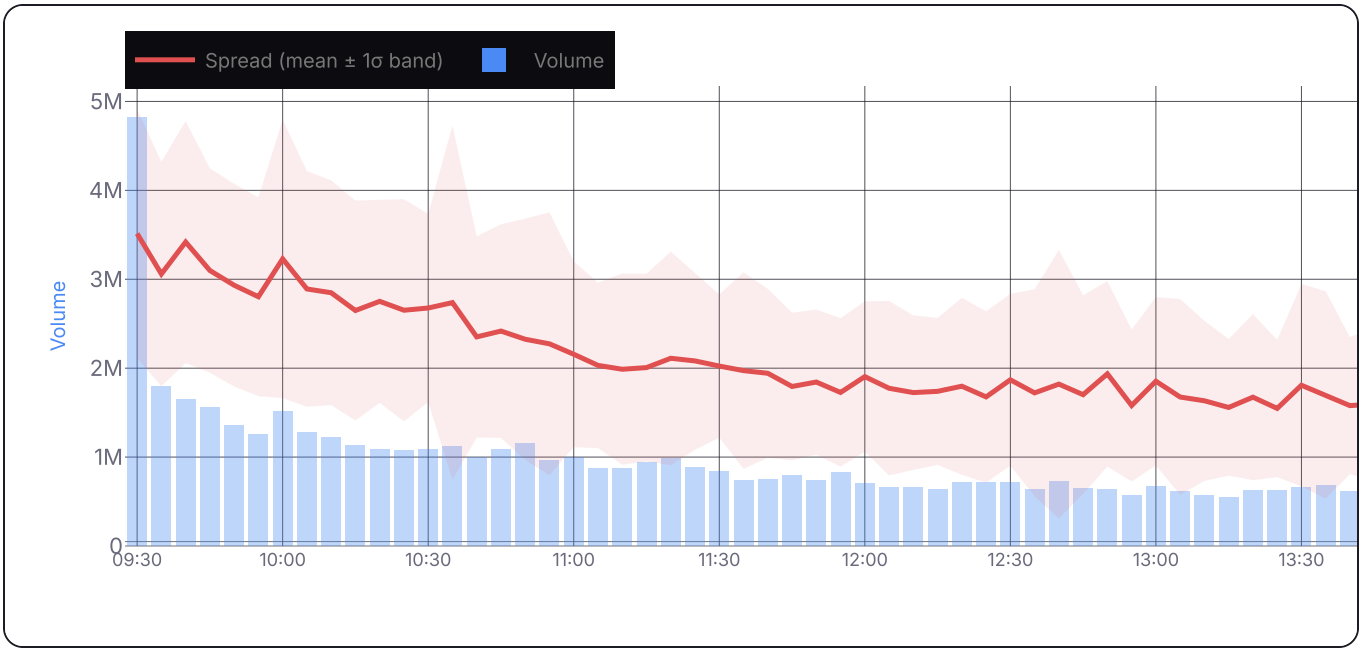
## Q2: Is it always equally cheap to trade?

Even when a market is open, the liquidity tax isn't constant. We measure it with the spread proxy — the gap between the highest and lowest price in each bar, in basis points. A wider spread means a higher tax on every trade.

SPY AAPL TSLA GME

### 2.1 — How much does the hidden cost change throughout the day?

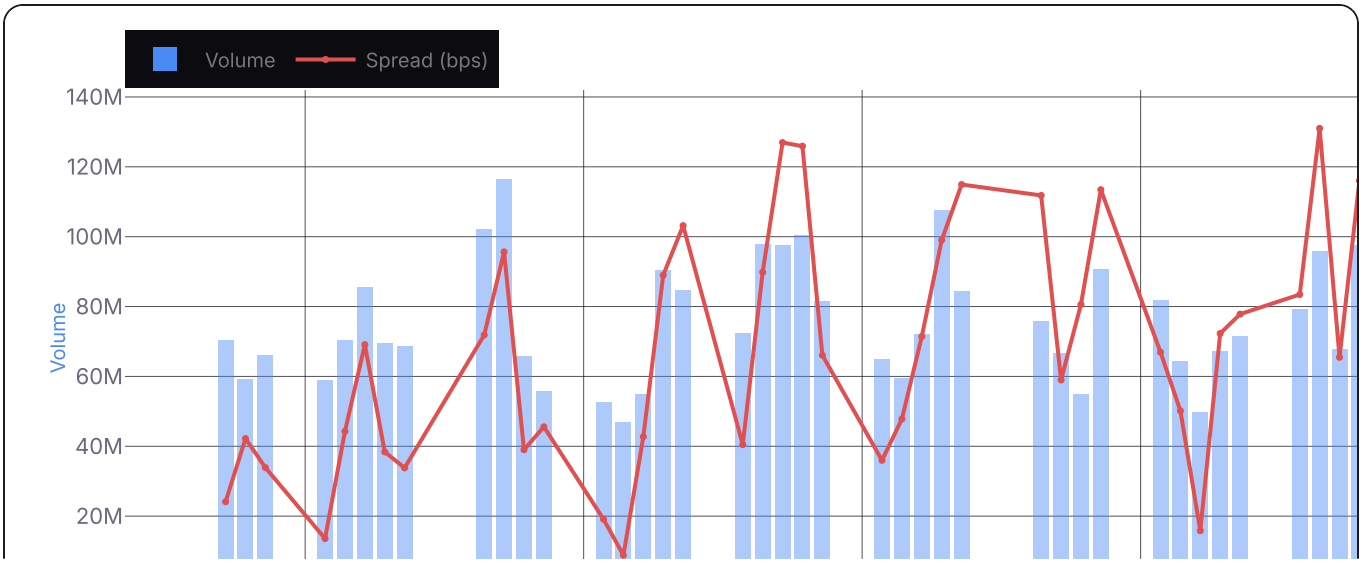
The chart below overlays average volume and average spread at 5-minute intervals across 60 trading days. Watch how the red spread line rises exactly when the blue volume drops.

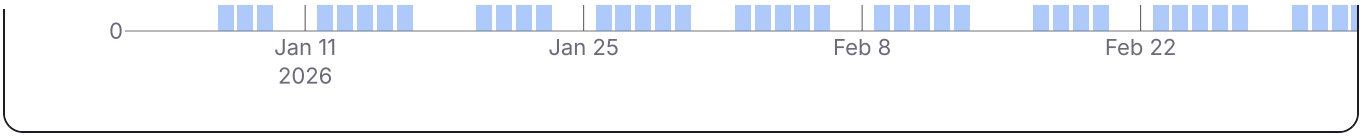



**WHY DUAL AXES?** Dual axes are controversial in visualization (Munzner, *Visualization Analysis and Design*, Ch. 5), but justified here: both series share a meaningful common dimension (time) and the visual task is detecting anti-correlation, not reading precise values. The shaded ±1σ band (Bollinger-style) adds uncertainty context.

### 2.2 — Are some hours 2x more expensive, or 10x?

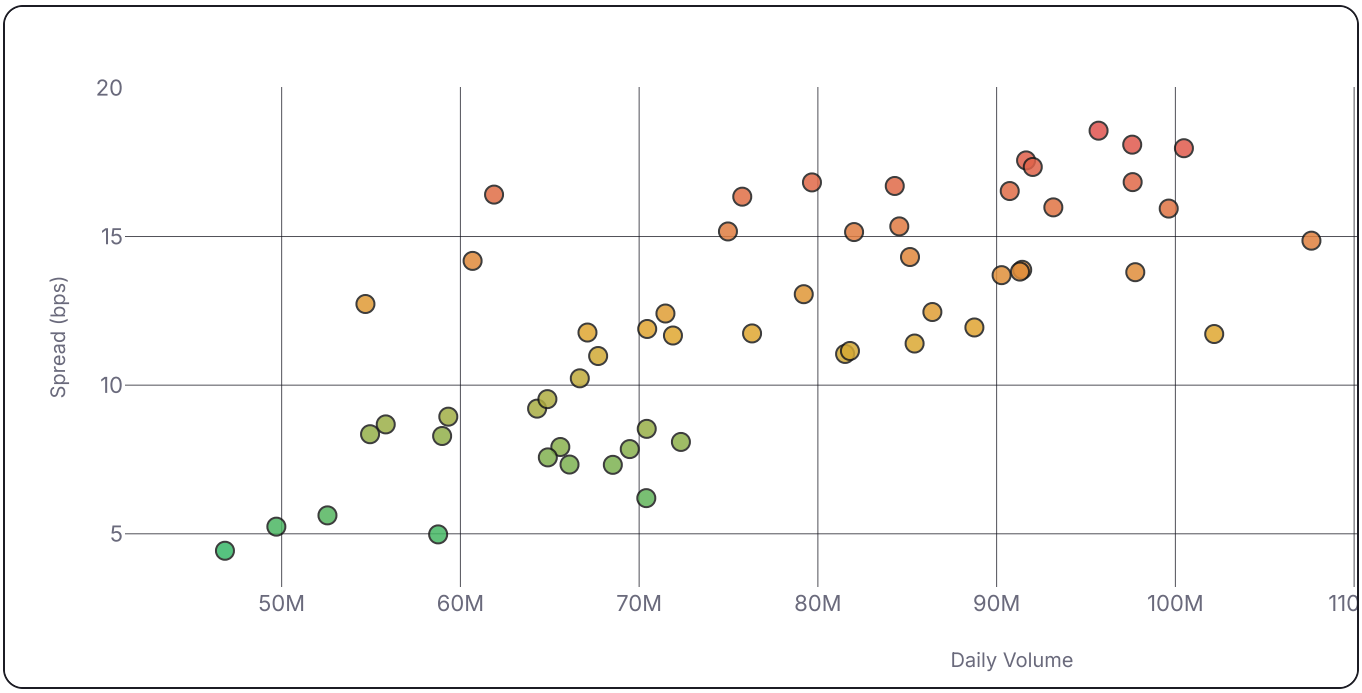
The daily view shows how spread varies day to day. Some days the cost is barely noticeable; on others it spikes to multiples of the average.






### 2.3 — If you could only trade at one time each day, when should it be?

The scatter plot below shows each trading day as a dot — volume on the X-axis, spread on the Y-axis. Color encodes spread (green = cheap, red = expensive). The answer is clear: trade when volume is highest and the dots cluster green.




> ***Answer:*** *For US equities, the first and last 30 minutes of the trading day are where spreads are tightest. The midday lull (12:00–14:00 ET) is measurably more expensive. For SPY, the difference can be 2–3x between peak and trough hours.*

## Q3: When lots of people are trading, does that make it cheaper?

This is the most common misconception about markets. Intuition says: more buyers and sellers = easier to trade = lower costs. The data says: not always.

SPY AAPL TSLA GME

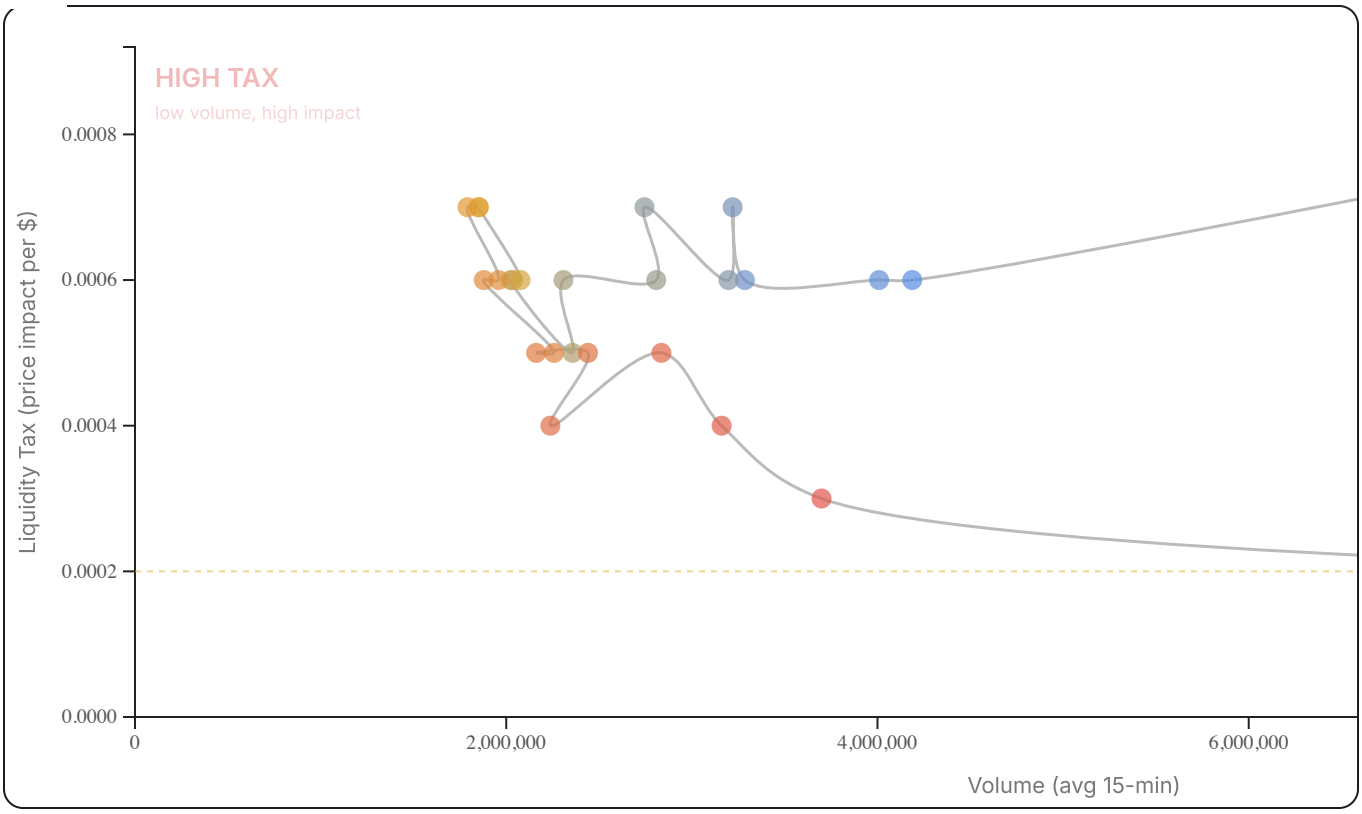
Every time you trade, you pay a hidden fee — the liquidity tax. It's the cost of your own trade pushing the price against you. Buy a stock and the price nudges up; sell and it nudges down. In liquid markets the tax is tiny. In illiquid ones it can eat your profits.

The animated chart below traces a single trading day, 15 minutes at a time. The horizontal axis shows how much trading is happening (volume). The vertical axis shows the liquidity tax — how much the price moves per dollar traded (the Amihud ratio).

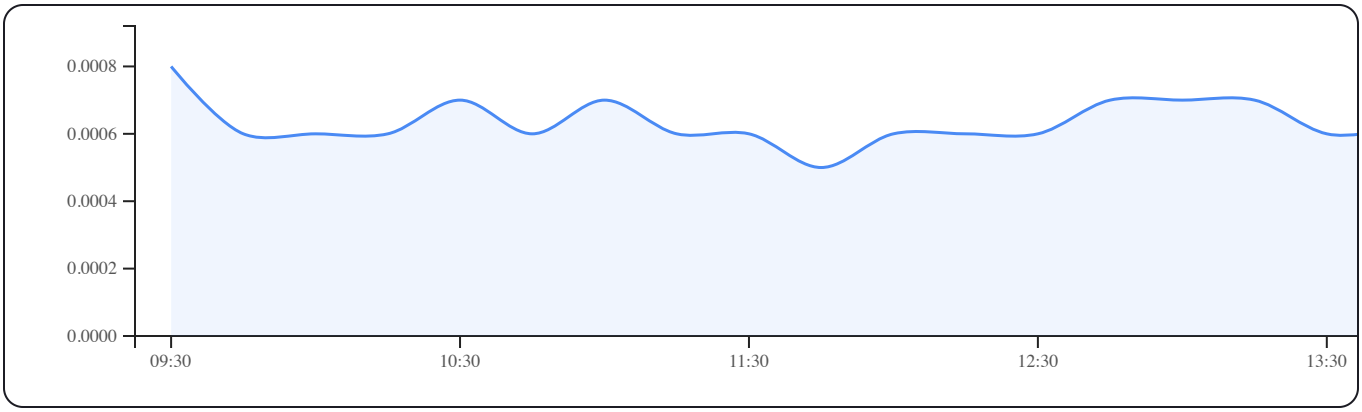
Bottom-right = best: high volume, low price impact — cheap to trade.
Top-left = worst: low volume, high price impact — expensive to trade.
Watch the dot trace through the day and see when the tax is lowest.

SPY AAPL TSLA GME

Play Reset



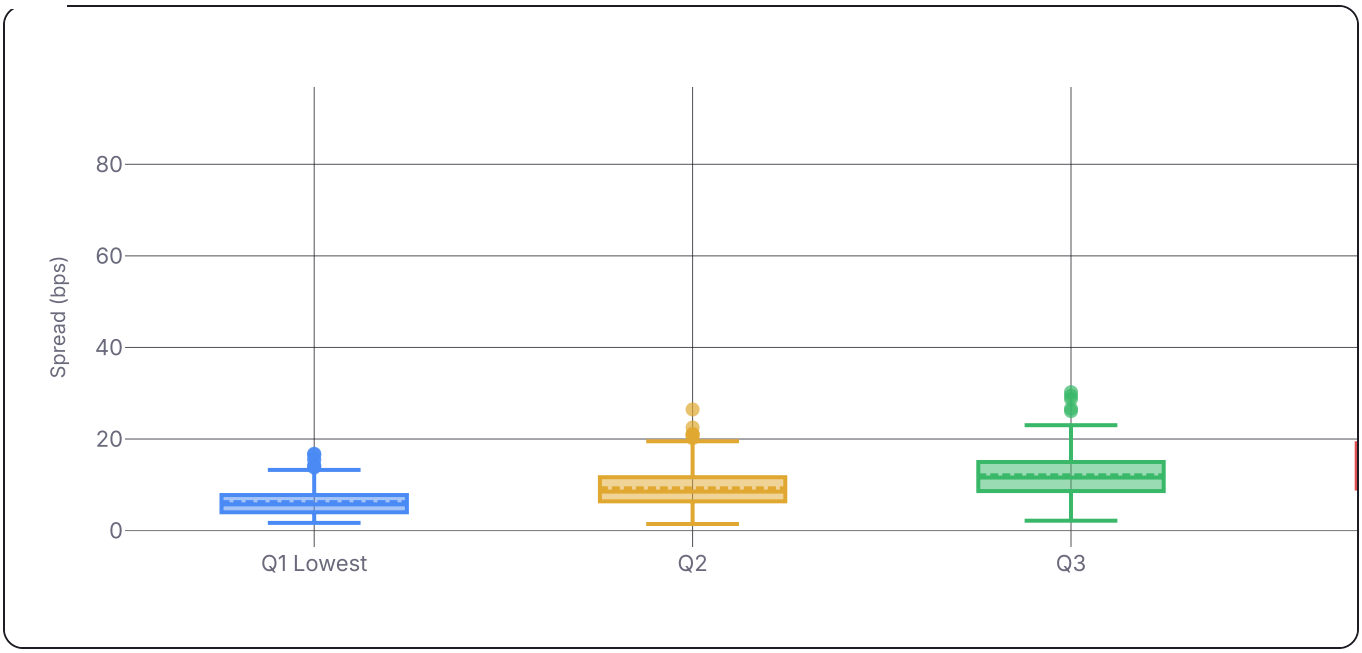

## LINKED TIME SERIES — HOVER OR CLICK A POINT ABOVE TO HIGHLIGHT THE TIME BELOW



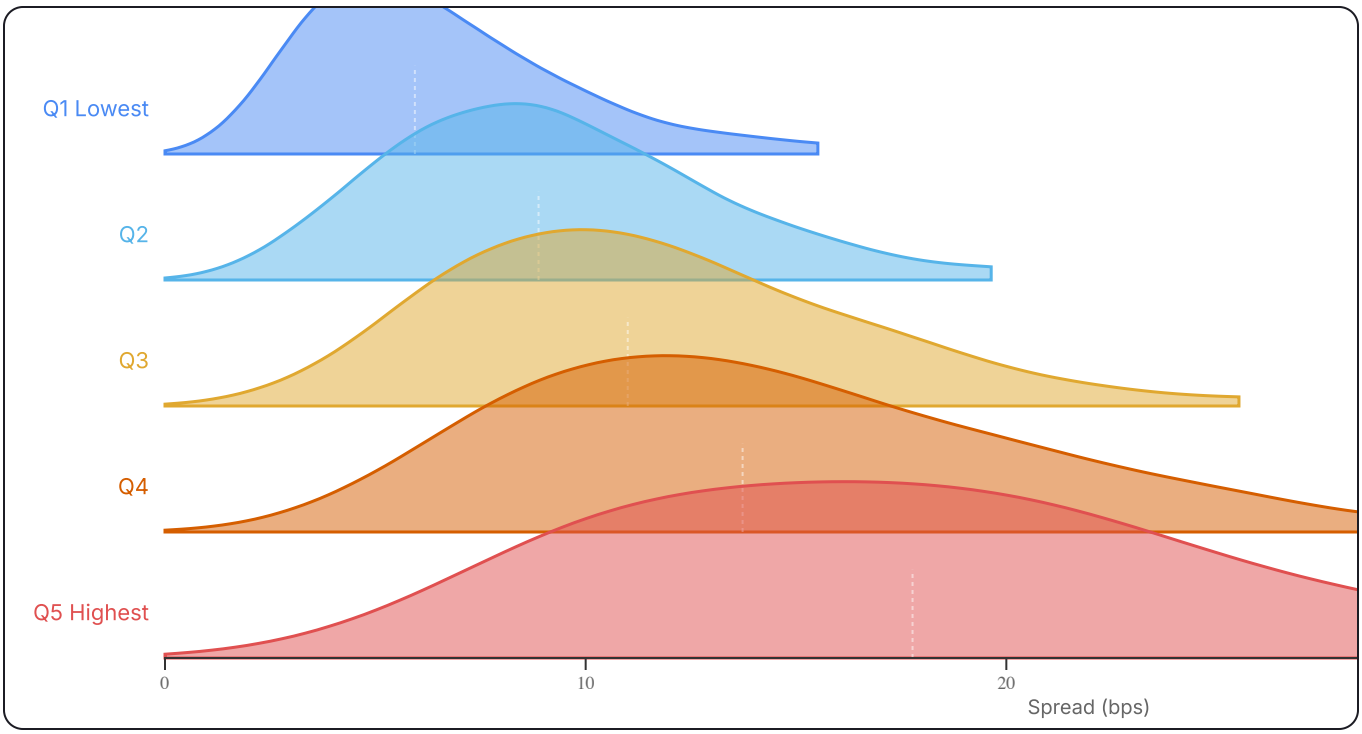

**WHY AN ANIMATED CONNECTED SCATTER?** Static scatter plots show correlation. The animated path adds *time* as a third dimension through motion (Rosling's "Gapminder" technique). The chronological trail reveals that volume and price impact don't just correlate — they follow a predictable daily arc. The linked time series below provides "details on demand" (Shneiderman's information-seeking mantra: overview first, zoom and filter, then details on demand).

### 3.1 — Do busy days always mean lower costs?

The box plots below partition all 5-minute bars into volume quintiles. If more volume always meant cheaper trading, you'd see a clean staircase going down from left to right. Do you?




## RIDGELINE PLOT — FULL SPREAD DENSITY PER QUINTILE (D3.JS)




**WHY BOX PLOTS AND RIDGELINE?** The scatter shows the cloud; the box plot slices it into groups and reveals the *distribution* at each level. Look at Q5 (highest volume): the median may be lower, but the tails are often fatter — high volume can come with extreme outliers.

### 3.2 — Are there days where everyone is trading but costs go up?

The rolling correlation below tracks how the volume-spread relationship changes over time. When the line is above zero, volume and spread are moving *together* — more trading, higher cost. When it's below zero, the normal relationship holds.

| +0.721 | Positive |
|---|---|
| OVERALL CORRELATION | MORE VOLUME = WIDER SPREADS |

### 3.3 — How often does the "more activity = cheaper" rule actually break?

Look at the correlation chart above. Every time the line crosses above zero, the rule is broken. For volatile stocks like TSLA and GME, this happens frequently. For SPY, it's rarer — but when it happens, it tends to be dramatic.

> ***Answer:*** *The relationship between volume and cost is unstable. On calm days, more volume does mean tighter spreads. But during stress, the correlation inverts: volume surges and spreads blow up simultaneously. The rule breaks exactly when you need it most.*

## Q4: Are some things just harder to trade than others?

Not all markets are created equal. Some assets are easy to buy and sell at any hour; others have wide spreads, thin volume, and unpredictable costs. Liquidity has multiple dimensions — let's map them.

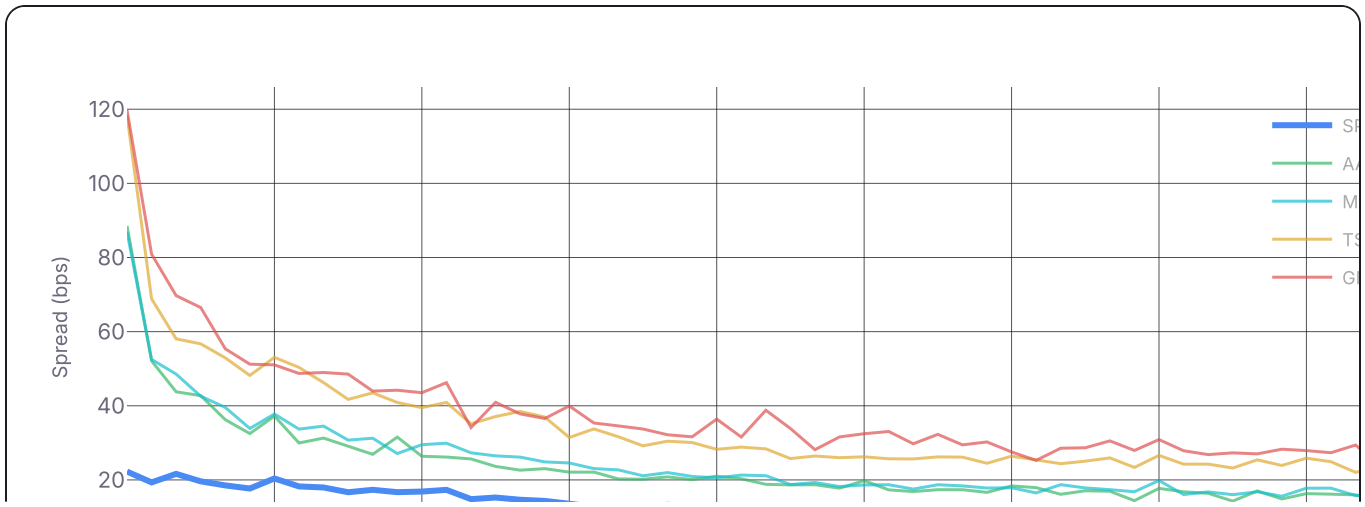
### 4.0 — Are ETFs easier to trade than individual stocks?

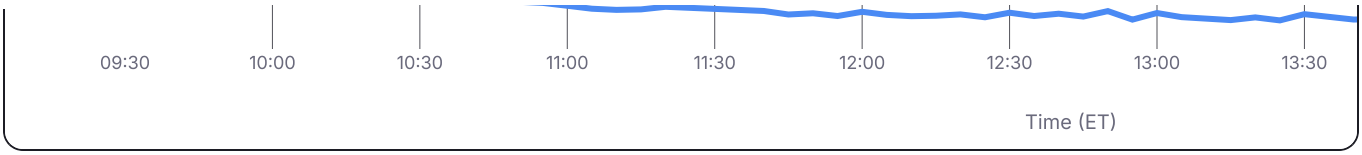
You might think a big stock like Apple is just as liquid as an ETF like SPY. The data shows otherwise. SPY holds the same stocks but is dramatically cheaper to trade because it has more market makers, tighter competition, and continuous arbitrage keeping spreads tight.

| 12.35 | 21.7 | 22.16 | 30.9 |
|---|---|---|---|
| SPY (S&P 500 ETF) AVG SPREAD (BPS) | AAPL (LARGE-CAP) AVG SPREAD (BPS) | MSFT (LARGE-CAP) AVG SPREAD (BPS) | TSLA (HIGH-VOL) AVG SPREAD (BPS) |

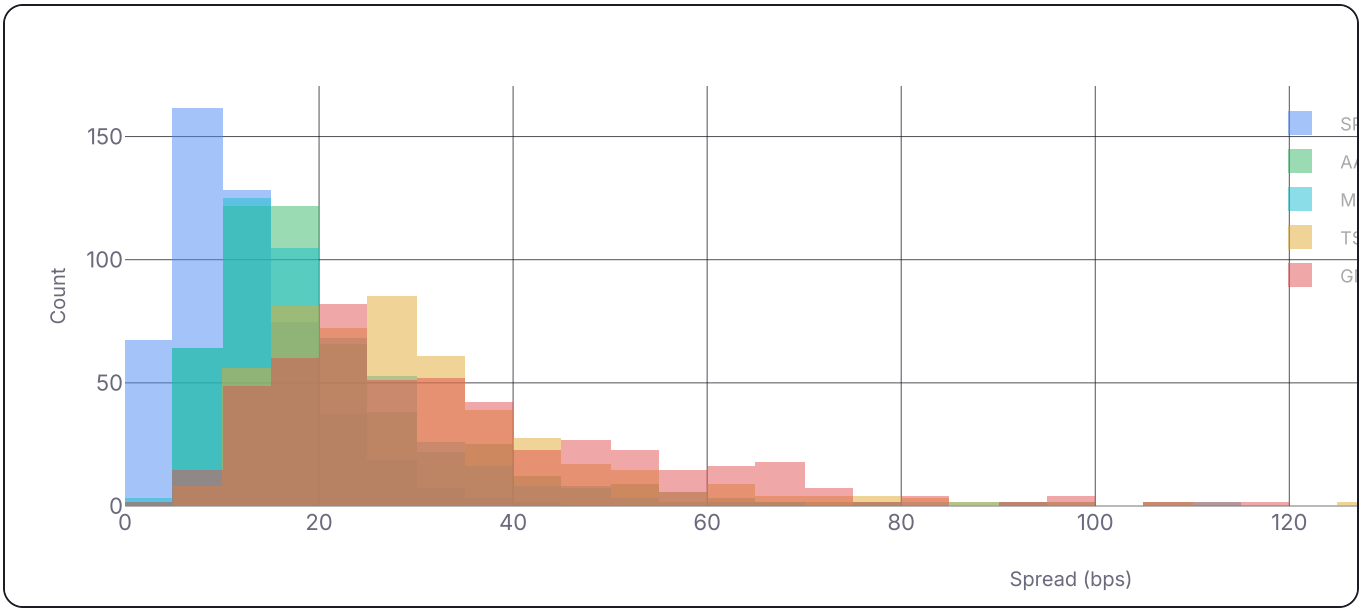
| 35.39 |
|---|
| GME (MEME STOCK) AVG SPREAD (BPS) |

#### INTRADAY SPREAD PROFILE — ALL FIVE SIDE BY SIDE






## SPREAD DISTRIBUTION — OVERLAID HISTOGRAMS

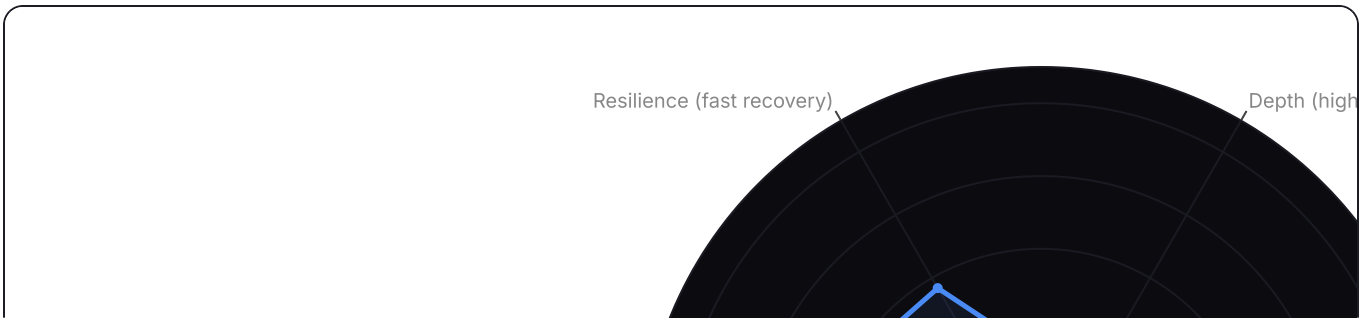


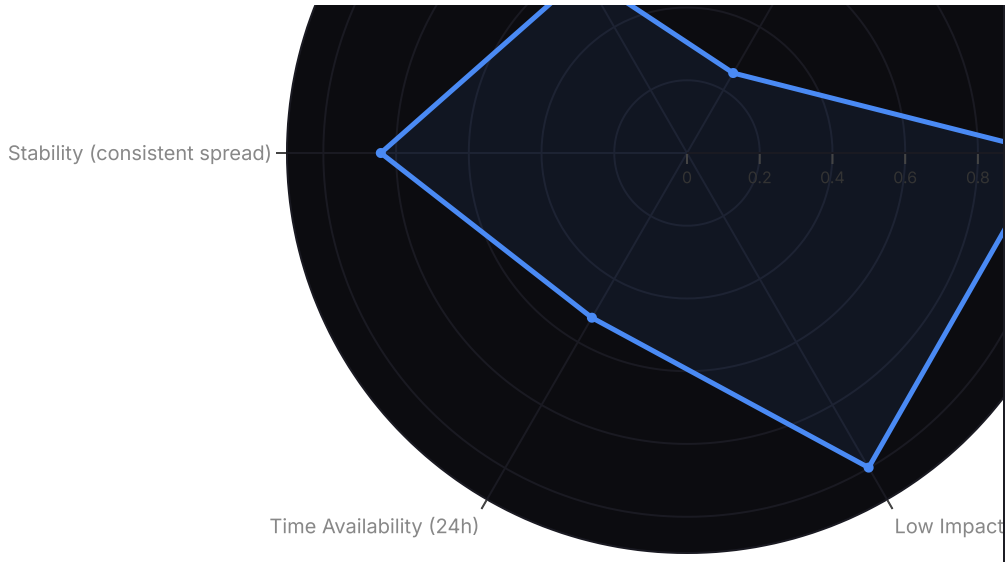

**WHY COMPARE ETFS TO STOCKS?** ETFs are often *more* liquid than the underlying stocks they contain. SPY holds all 500 S&P stocks, yet its spread is tighter than any individual component. This is counterintuitive — how can a basket be cheaper to trade than its parts? The answer is market structure: ETFs have dedicated market makers and arbitrage mechanisms that individual stocks lack. This comparison makes the concept of structural liquidity tangible.

### 4.1 — What makes gold different from wheat from a trading cost perspective?

The radar chart below scores each asset on six dimensions of liquidity. Click up to 3 assets to overlay and compare their "fingerprints." Bigger shape = more liquid across the board.

EUR/USD | GBP/USD | S&P 500 | Nikkei | FTSE | Gold | Oil | US Bond | BTC | Wheat | Nat Gas | Corn



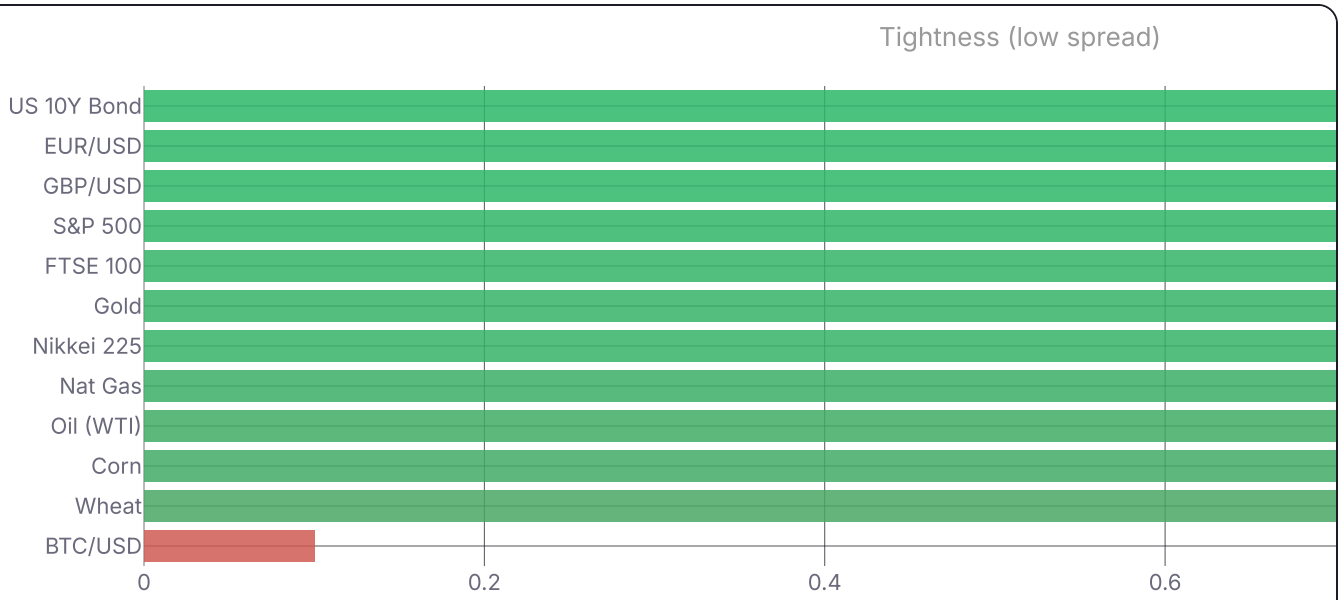



**WHY A RADAR CHART?** Radar charts get criticized in visualization, but here the shape *is* the message. A large, round polygon = uniformly liquid. A small, spiky one = uneven and illiquid. The gestalt comparison — EUR/USD's expansive circle vs Wheat's collapsed triangle — is immediately readable without studying any axis individually.

### 4.2 — Is the gap between "easy" and "hard" assets small or enormous?

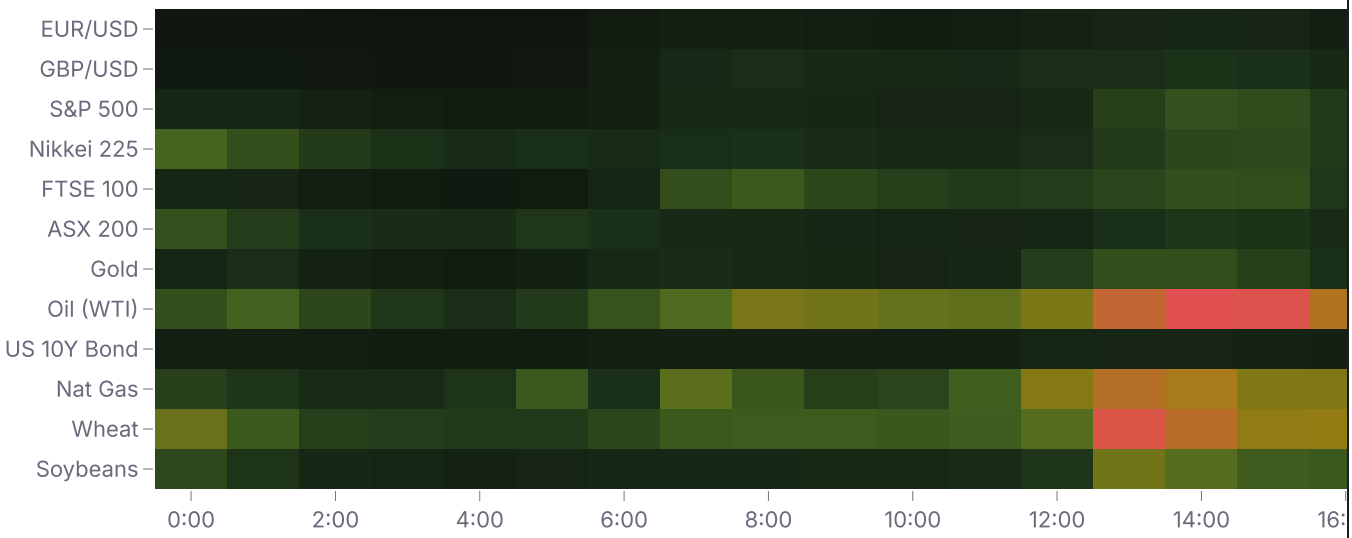
Pick a dimension below to see the full ranking. The gap between first and last isn't a subtle difference — it's often an order of magnitude.

Tightness | Depth | Resilience | Stability | Time Availability | Low Impact



### 4.3 — Do cheap-to-trade assets stay cheap at all hours, or do they have weak spots?

The heatmap below shows the actual spread for each asset at every hour of the day. Even EUR/USD — the most liquid market in the world — has hours where its spread widens noticeably. Every asset has a "liquidity geography."




**WHY A HEATMAP HERE?** The radar chart compares assets on abstract, averaged dimensions. The heatmap grounds it in operational reality: at 3am UTC, what does it *actually* cost to trade each asset? Color intensity answers that question at a glance for every asset and every hour.

> ***Answer:*** *The gap is enormous. EUR/USD's spread is often 50–100x tighter than Wheat's. But even liquid assets have weak spots: Gold widens at night, the S&P 500 thins out on weekends. "Liquid" is not a permanent state — it's a function of what, when, and where.*

## Q5: What happens to trading costs when markets panic?

On March 9, 2020, the S&P 500 hit a circuit breaker for the first time in over a decade. COVID-19 fears collided with an oil price war. What happened to the liquidity tax?

### 5.1 — How much more expensive does it get on a crash day vs a normal day?

The timeline below shows price and spread over four months. The spread bars are colored by severity: green = normal, amber = elevated, red = extreme.

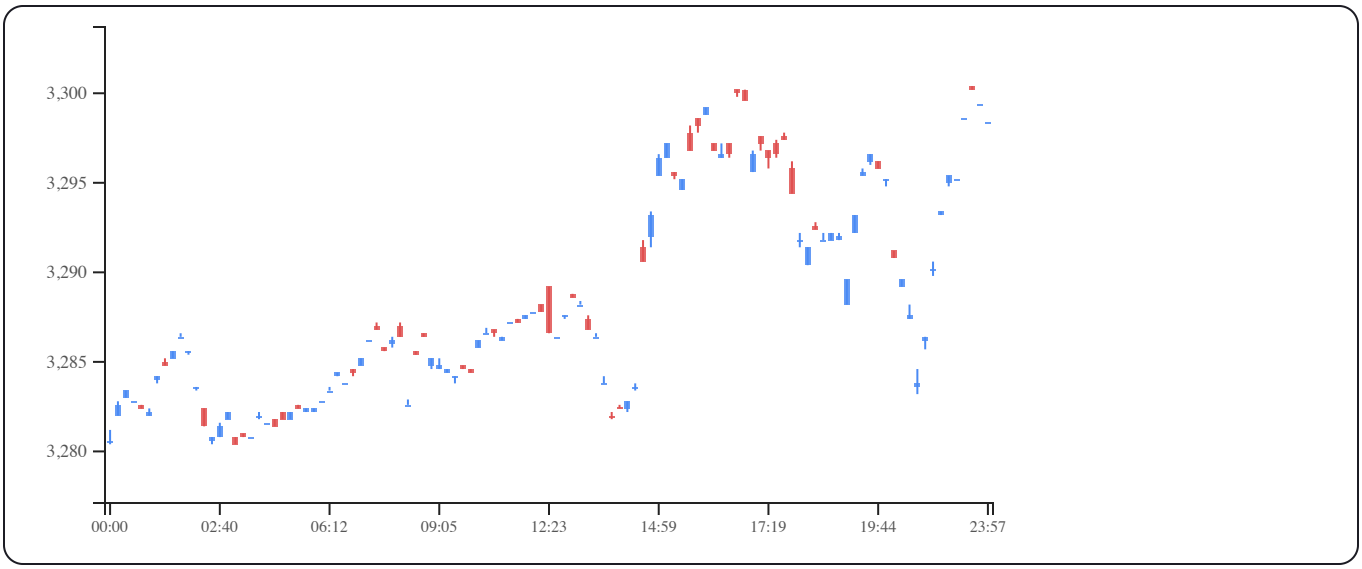
| $0.4 | $4.7 | 11.8x |
|---|---|---|
| NORMAL MEDIAN SPREAD | CRASH MEDIAN SPREAD | SPREAD MULTIPLIER |

| $40.7 | $535.7 | $3651.4 |
|---|---|---|
| CRASH MAX SPREAD | NORMAL DAY TOTAL COST | CRASH DAY TOTAL COST |

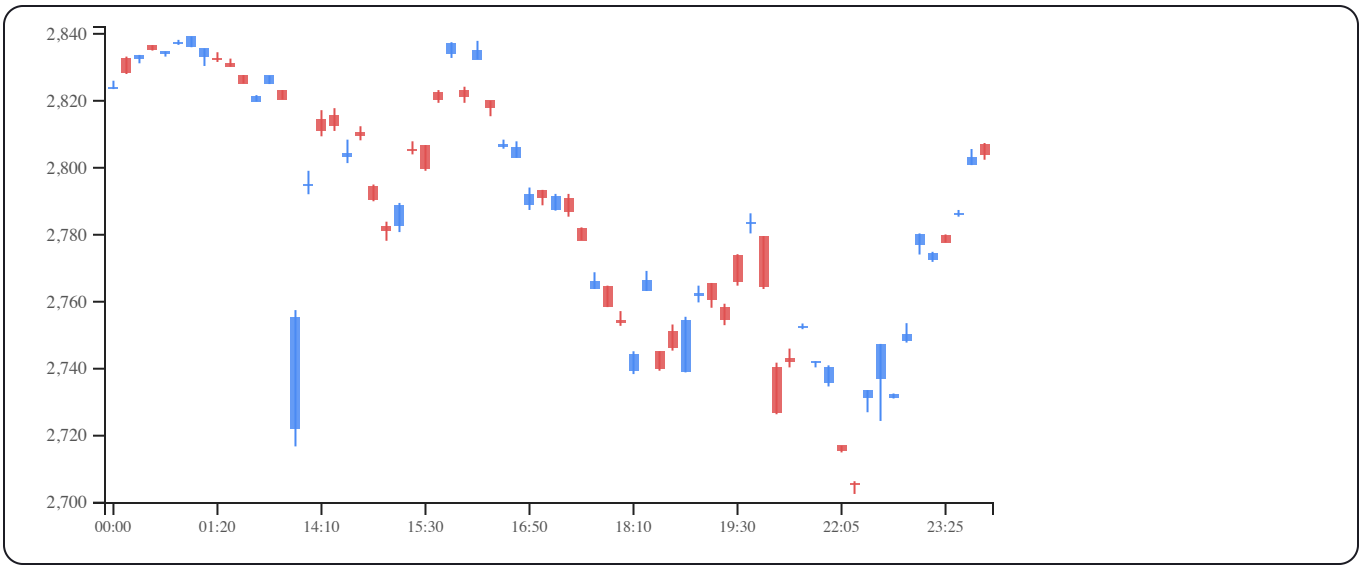
Now zoom into two single days. **Both charts share the same Y-axis** so the comparison is honest — the crash day spreads aren't "a bit wider," they're in a completely different range.

## CANDLESTICK CHARTS (D3.JS) — OHLC PRICE ACTION, MINUTE BY MINUTE

January 15, 2020 — Normal Day




March 9, 2020 — Crash Day

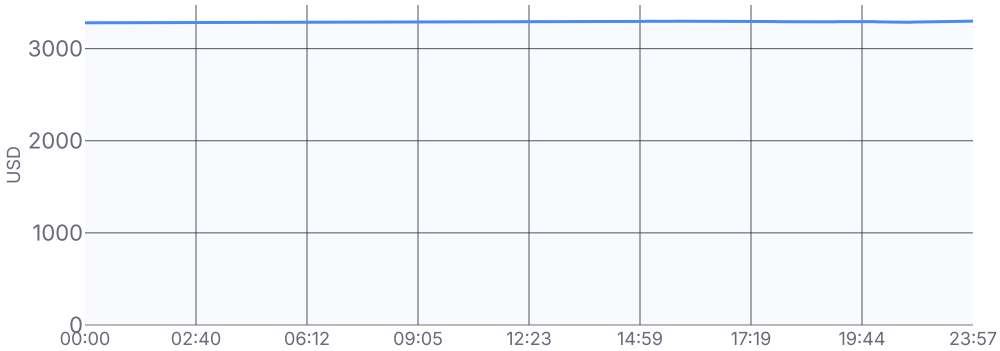



**WHY CANDLESTICKS?** The candlestick encodes four values per bar (open, high, low, close) — a remarkably data-dense representation developed in 18th-century Japan. Color encodes direction (blue = up, red = down). The wicks show intrabar range — which is our spread proxy. On the crash

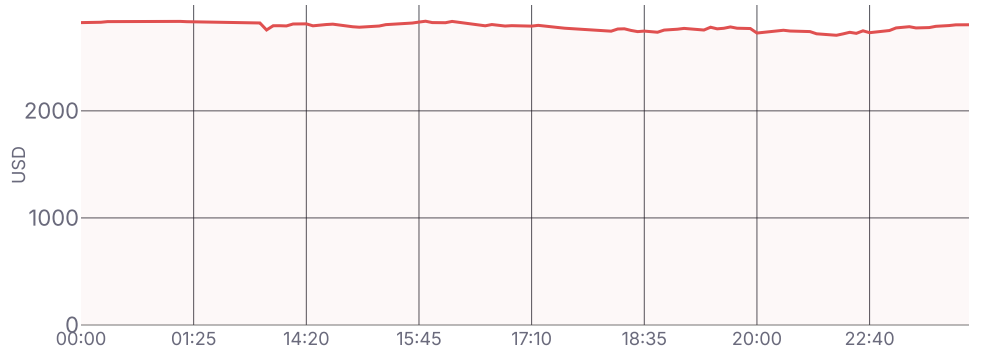
day, notice how the wicks are dramatically longer and the bodies smaller (price reverses within bars). Custom D3.js implementation.

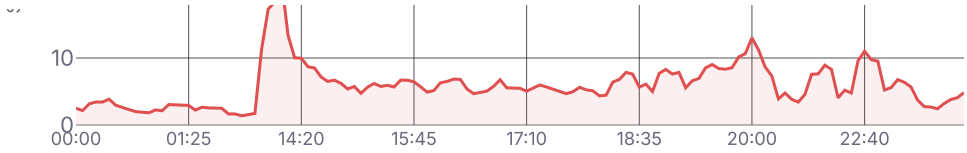
## SPREAD EVOLUTION — SMOOTHED MINUTE-BY-MINUTE (SHARED Y-AXIS)

### Normal — Price & Spread



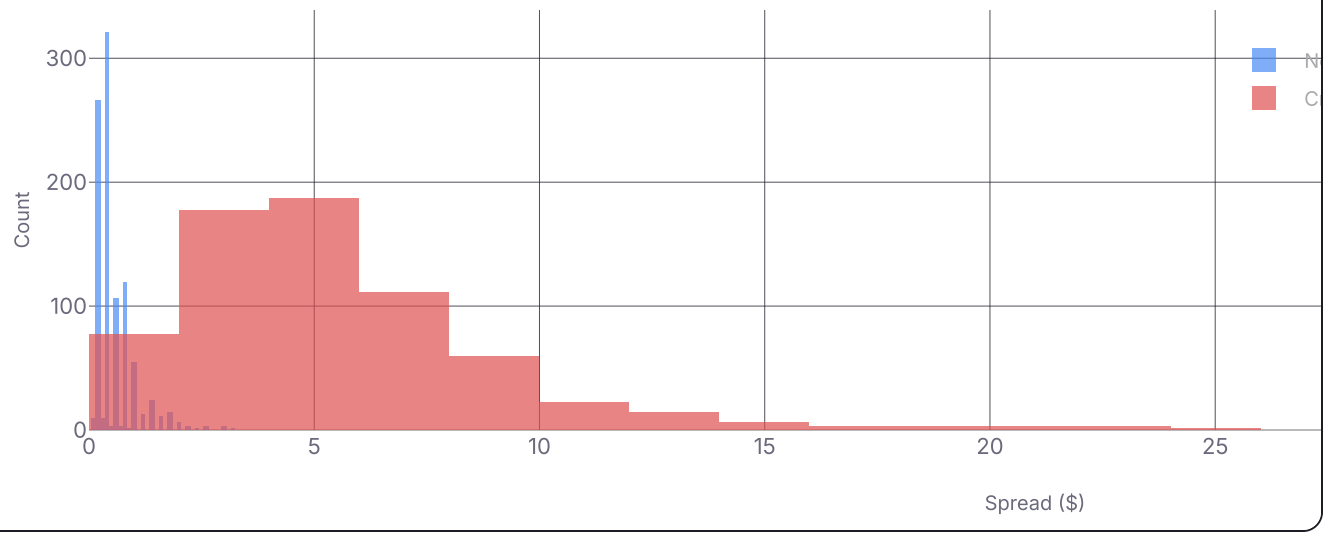
### Crash — Price & Spread






### 5.2 — Does the cost spike happen instantly or build up gradually?

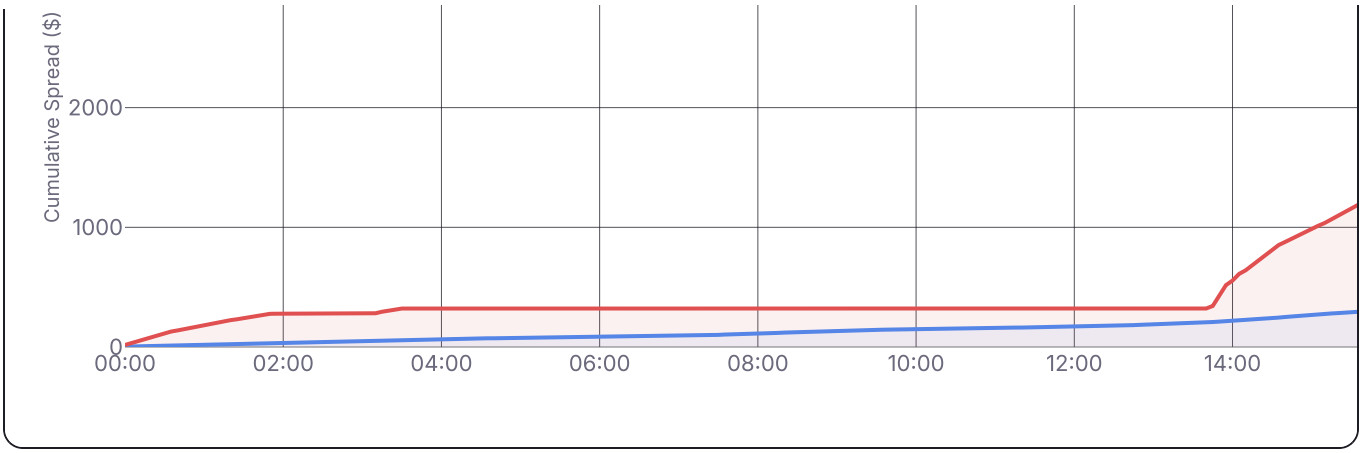
The overlaid histograms show the full distribution of minute-by-minute spreads for each day. On the normal day, most spreads cluster near zero. On the crash day, the entire distribution shifts right — and develops a long, fat tail of extreme values.



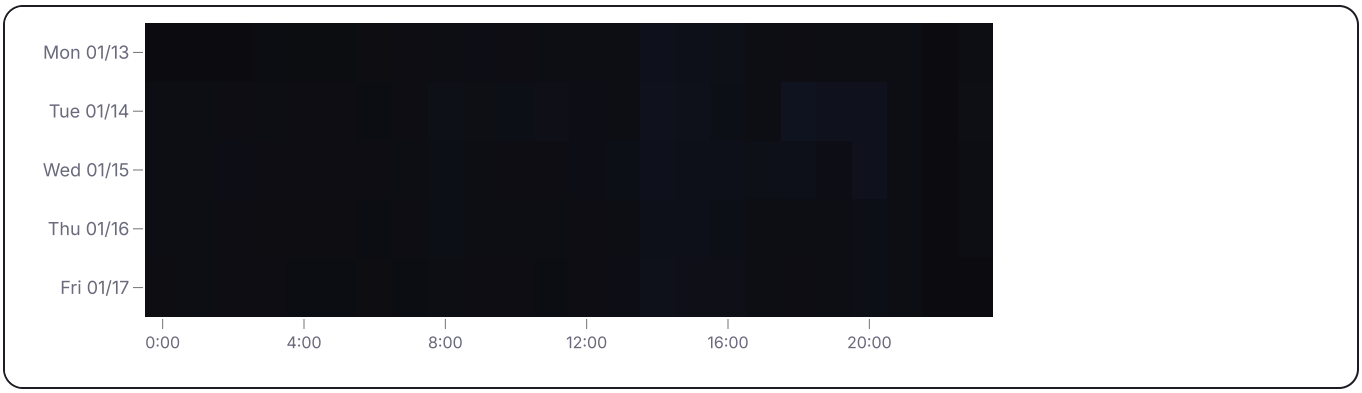

**WHY OVERLAY HISTOGRAMS?** Side-by-side time series show *when* spreads differ. The histogram shows *by how much*. The normal day's distribution barely extends past where the crash day's distribution *begins*. This is the clearest way to show the magnitude of the shift.

### 5.3 — How long does it take for things to go back to normal?

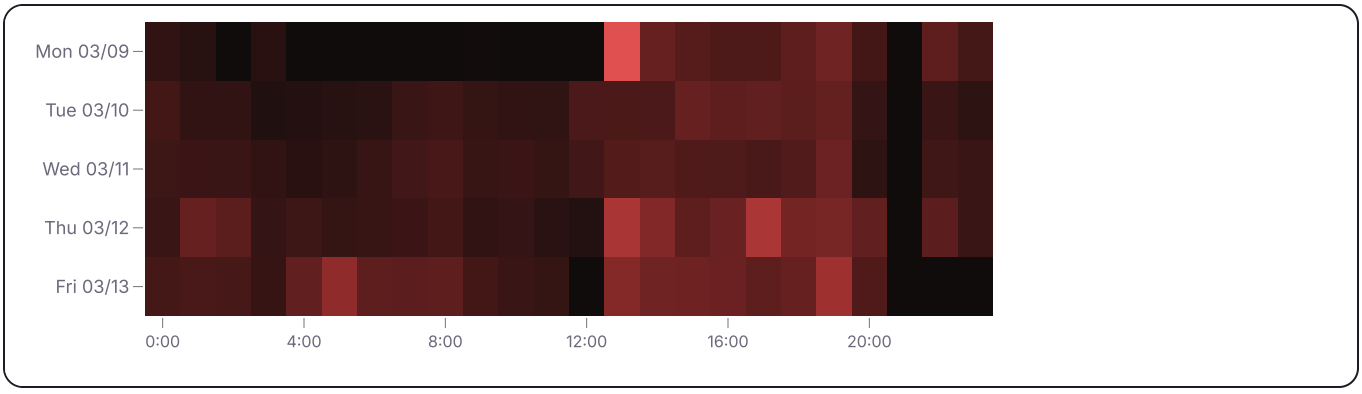
The cumulative chart shows the total liquidity tax accumulating throughout each day. By close, the crash day's total cost dwarfs the normal day's. And the weekly heatmaps below show it took the entire week of March 9–13 for spreads to begin recovering — while the normal week is uniformly calm.




## Normal Week (Jan 13–17)




## Crash Week (Mar 9–13)




**WHY SO MANY VIEWS OF THE SAME EVENT?** This "coordinated multiple views" approach (Roberts, 2007; Munzner Ch. 12) uses different chart types to answer different sub-questions about the same phenomenon. Timeline → *when*. Candlestick → *market microstructure*. Day comparison → *visual contrast*. Histogram → *severity*. Cumulative → *total cost*. Heatmap → *recovery speed*. No single chart could carry all of these answers.

> ***Answer:*** *Trading costs explode instantly — within minutes of the crash — but recovery is slow and gradual. The asymmetry is striking: liquidity disappears in seconds but takes days to fully return. This is* ***liquidity evaporation****, and it's the reason "you can't always trade when you want to."*

---

## So, why can't you always trade when you want to?

Because the liquidity tax — the hidden cost of every trade — is never constant. It shifts by the hour, by the asset, and by the state of the world. Now you know how to see it.

⏰

**Every market has a rhythm**

Stocks follow a U-shape. Currencies peak during session overlaps. Commodities have narrow windows. Trading outside peak hours costs more.

📈

**Not all assets are equal**

EUR/USD and major indices are deeply liquid. Commodities, small-caps, and crypto have wider spreads and higher impact — sometimes by 50x or more.

### The liquidity tax spikes in crises

During crises, volume surges but spreads explode. The liquidity tax skyrockets exactly when you want to trade most — and recovers slowly.

DATA SOURCES & METHODOLOGY

**Equity data** (AAPL, MSFT, SPY, TSLA, GME): Yahoo Finance via yfinance — 1-min and 5-min intraday bars.
**FX & Commodity data** (EUR/USD, SPX500, Gold, Oil, Wheat, Sugar, Corn, Nat Gas): Kaggle/Oanda minute-bar dataset (2005–2020).
**Crypto data** (BTC/USD): Bitstamp exchange data (2012–2018).

**Spread proxy:** (High − Low) / Close × 10,000, expressed in basis points. This approximates the bid-ask spread using publicly available OHLC data. It overstates the true spread (it captures intra-bar volatility), but the relative comparisons across assets and time periods remain valid.

**Liquidity dimensions** (radar chart): Tightness = inverse median spread; Depth = median daily volume; Resilience = inverse of 5-lag spread autocorrelation; Stability = inverse of spread CV; Time Availability = evenness of hourly volume; Impact = inverse of return-per-volume. All normalized to [0.1, 1.0] across the asset universe.

Liquidity is something you can see — if you know where to look.

## Appendix: The Full Picture

The visualizations above answered five questions. These heatmaps let you explore the data further — each one encodes thousands of data points into a single image. Read them like a weather map: color intensity reveals patterns that tables and line charts would hide.

### A.1 — Liquidity Heatmap: Hour × Day-of-Year

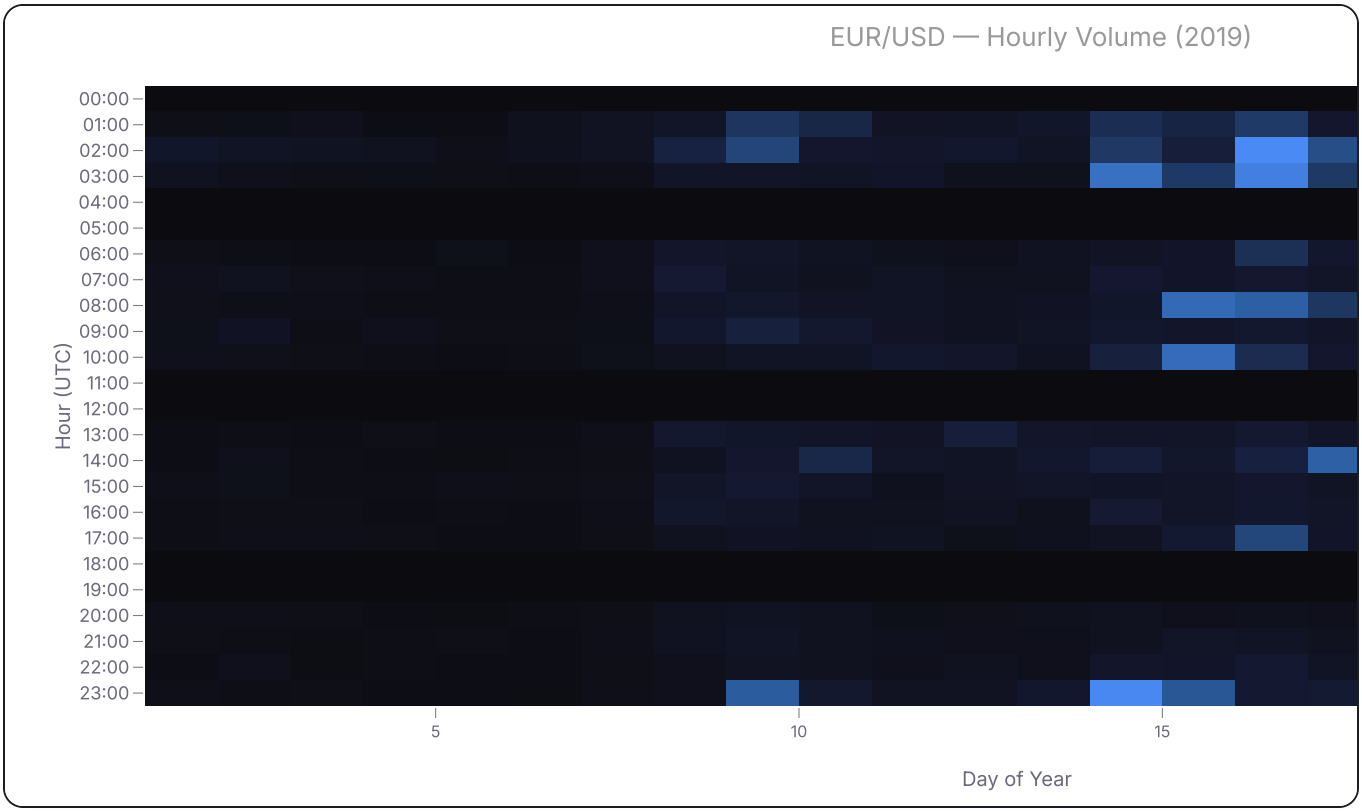
Each row is an hour of the day. Each column is a day of the year. Color = average spread (trading cost) at that hour on that day. Brighter = wider spread = more expensive.

EUR/USD | Gold | S&P 500 | US 10Y Bond | BTC/USD

### A.2 — Volume Heatmap: Hour × Day-of-Year

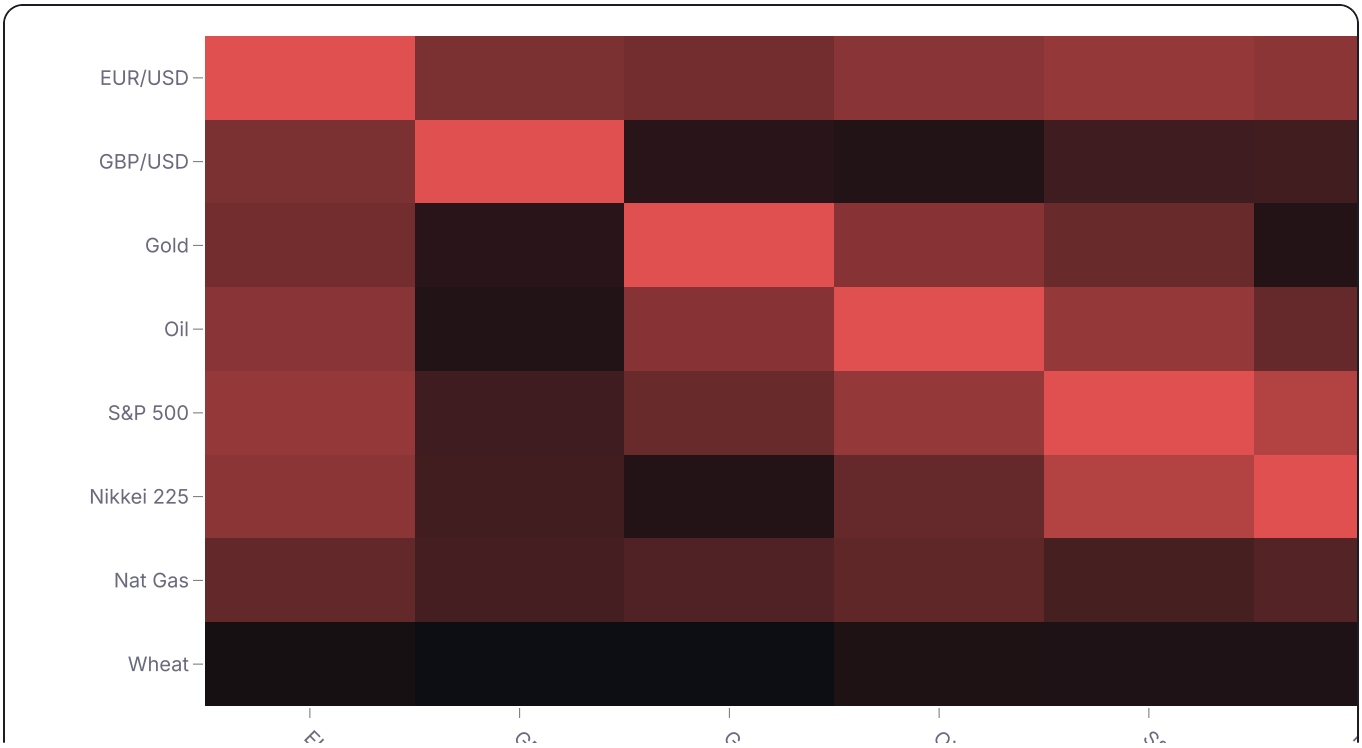
Same layout, but color = average volume. Compare with A.1 above — do the "hot" volume spots always align with the "cool" spread spots?

EUR/USD | Gold | S&P 500 | US 10Y Bond




## A.3 — Cross-Asset Liquidity Correlation

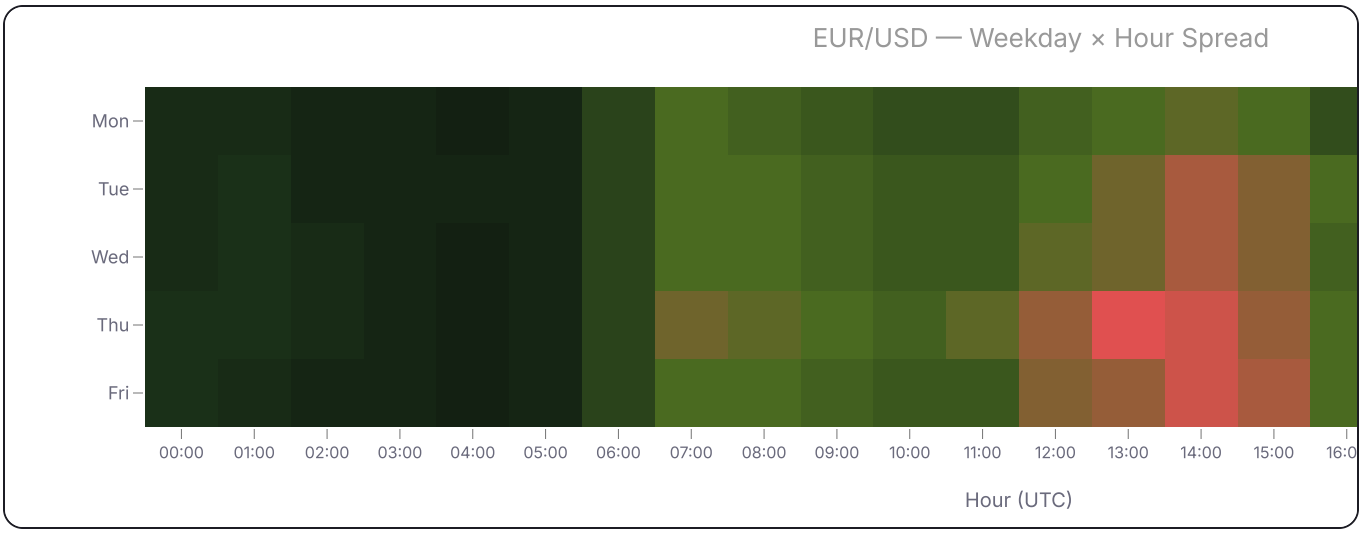
Do liquidity conditions move together across markets? This matrix shows the Pearson correlation of daily spreads between every pair of instruments. High correlation = when one market gets illiquid, so does the other.



## A.4 — Day-of-Week × Hour Spread Heatmap

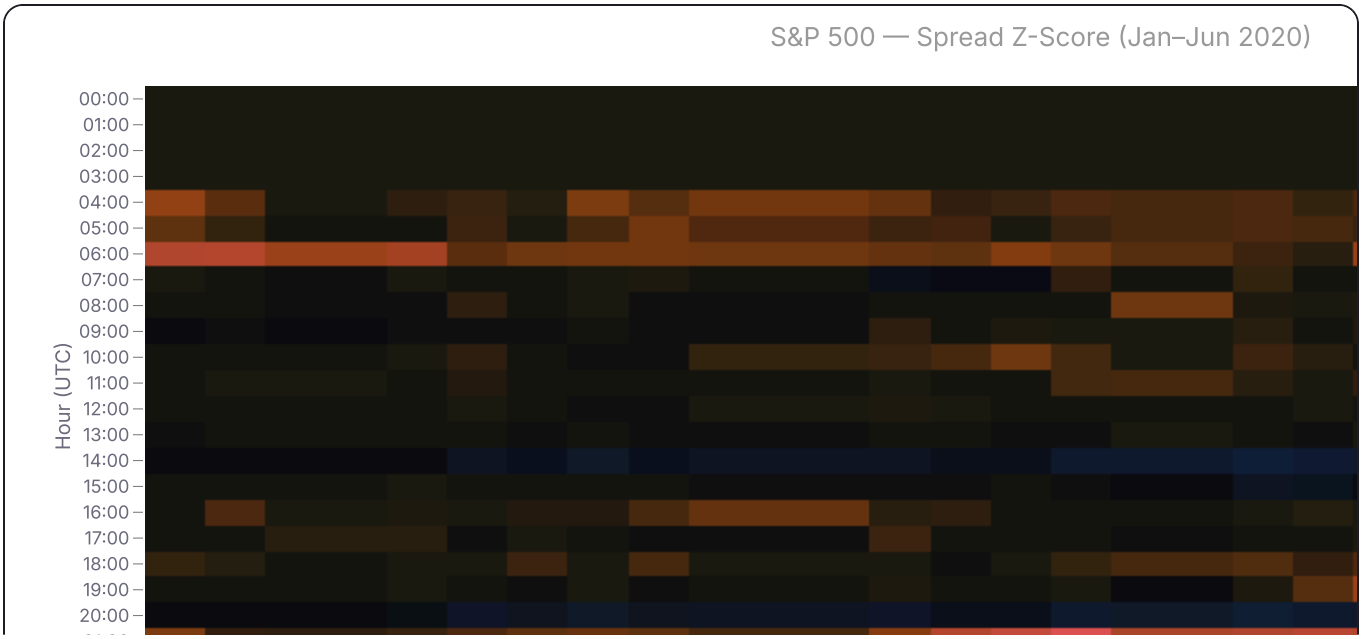
The weekly rhythm. Is Monday morning different from Friday afternoon?

EUR/USD | Gold | S&P 500 | US 10Y Bond | BTC/USD



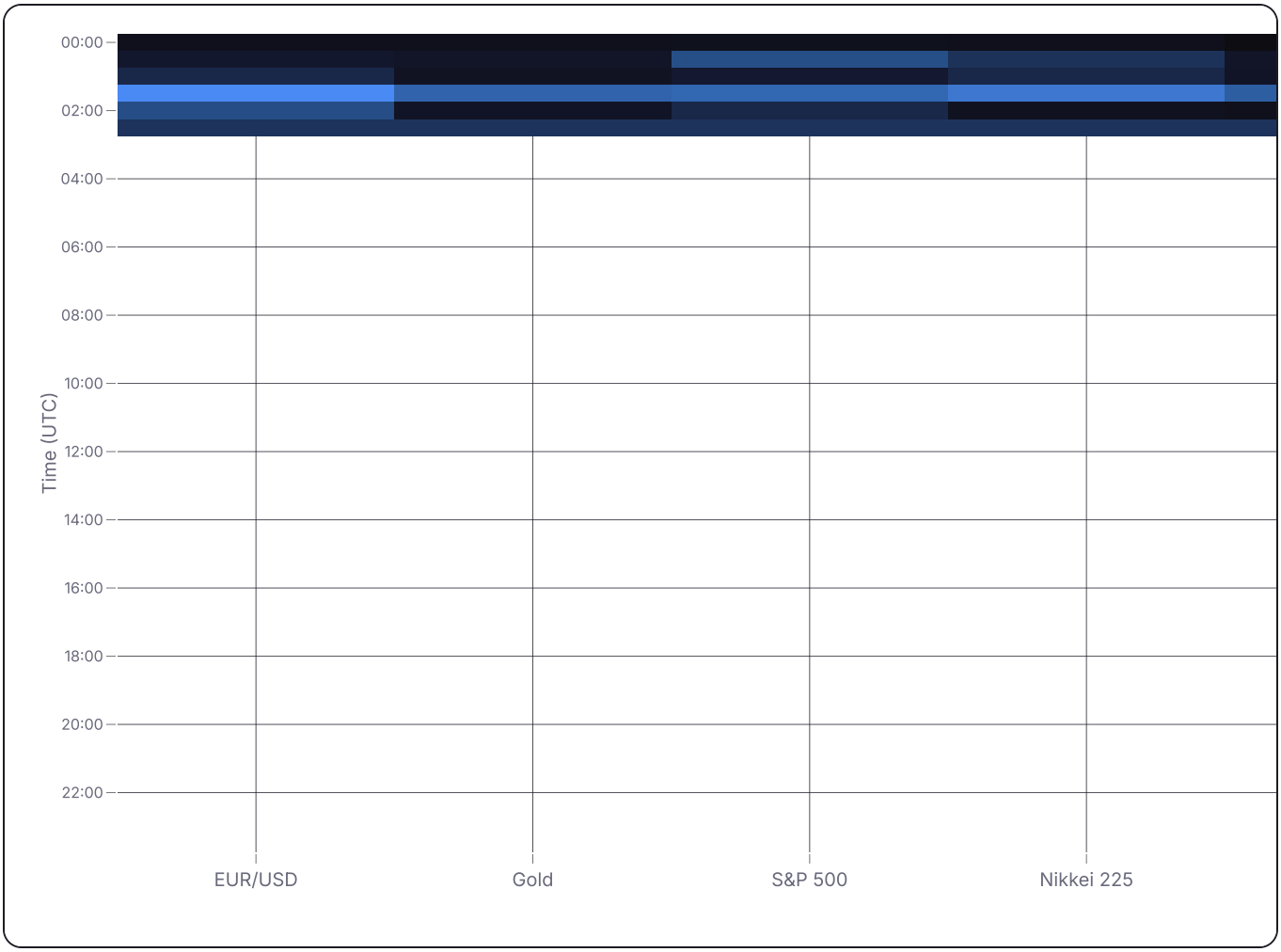

## A.5 — Liquidity Crisis Detector: Spread Z-Scores (S&P 500, Jan–Jun 2020)

Raw spreads have a baseline pattern (high at night, low during trading hours). Z-scores remove that pattern and highlight anomalies — hours where spreads were unusually wide relative to their recent history. The COVID crash in March should light up like a flare.



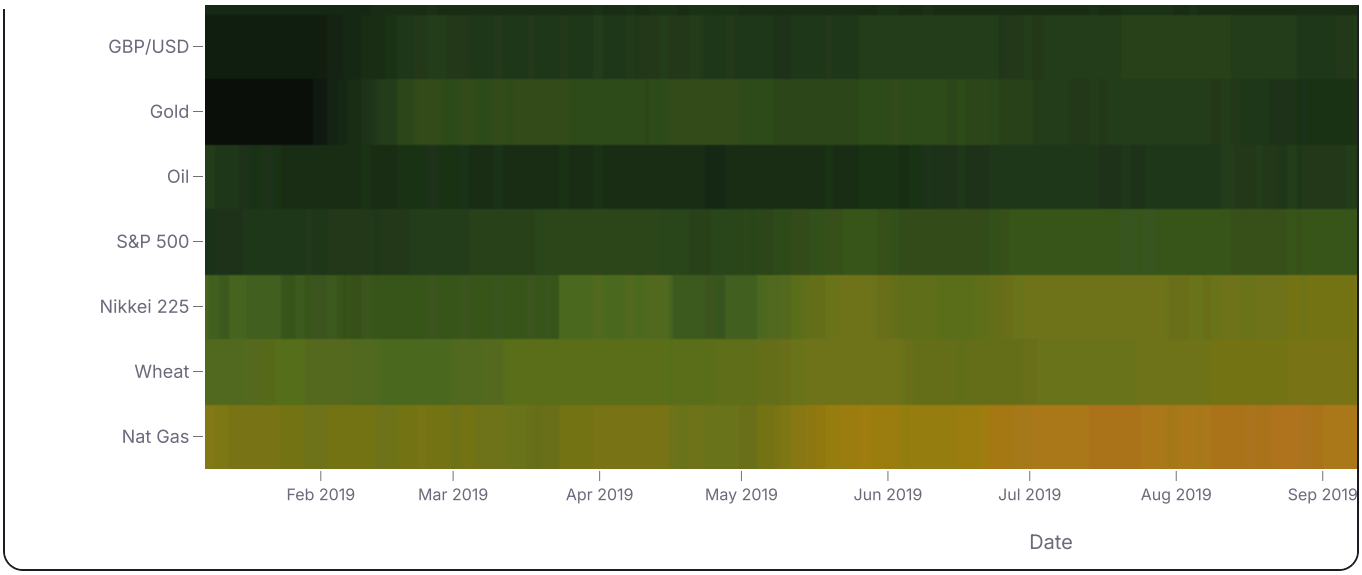
## A.6 — Volume Profile: When Does Each Market Trade?

Each column is an instrument. Each row is a 30-minute time slot. Color shows what percentage of the day's volume trades in that slot. A single view comparing the "heartbeat" of every market.




## A.7 — Rolling Liquidity Regime: Who Goes Illiquid When?

Each row is an instrument. Each column is a trading day in 2019. Color = rolling 20-day average of illiquidity (Amihud-like ratio, log scale). Brighter = more illiquid. Look for vertical stripes — those are days when multiple markets went illiquid simultaneously.



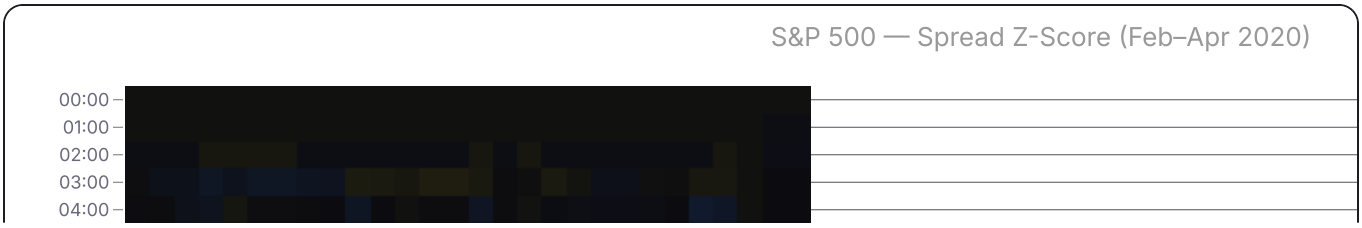

## Deep Dive: Anatomy of a Liquidity Crisis

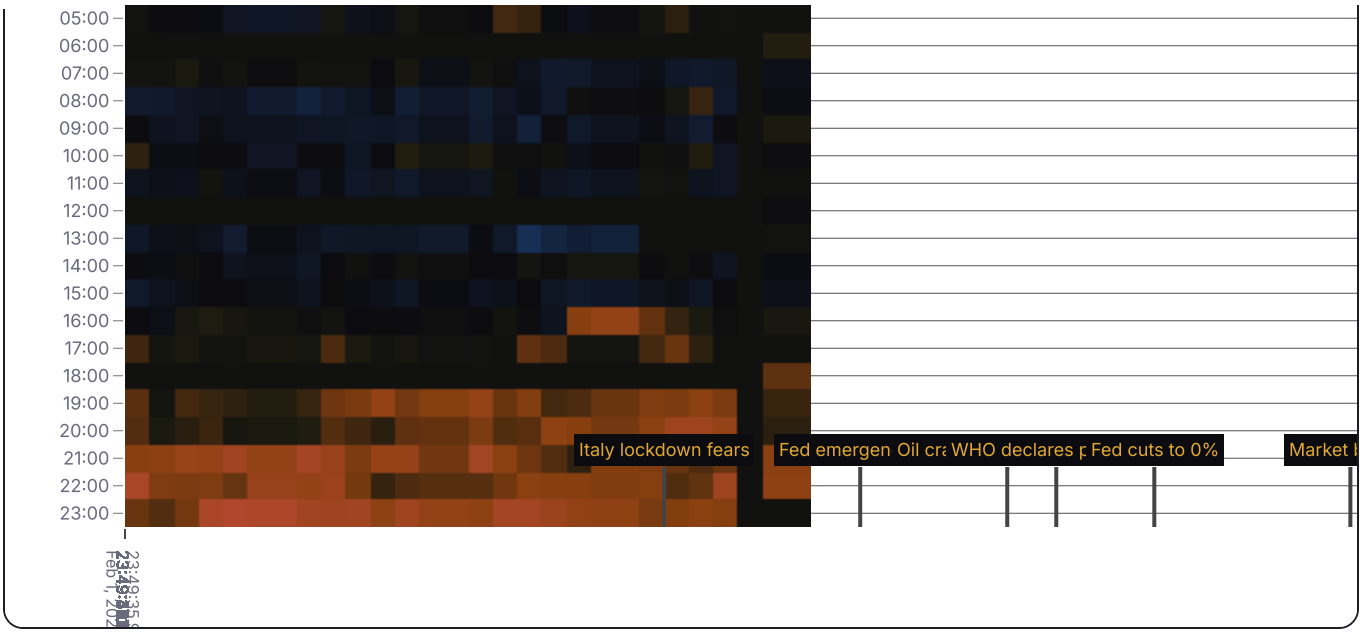
The heatmaps above showed the data. These three visualizations tell the *story* — how liquidity collapsed across the entire financial system in early 2020, and what the warning signs looked like.

### D.1 — The Crisis Unfolds: Multi-Asset Z-Scores (Feb–Apr 2020)

Each heatmap shows spread z-scores for one asset. Blue = unusually tight spreads. Red = unusually wide (liquidity stress). Key dates are annotated. Watch how stress appears first in one market, then spreads to all of them.

S&P 500 | EUR/USD | Gold | Oil | Nikkei



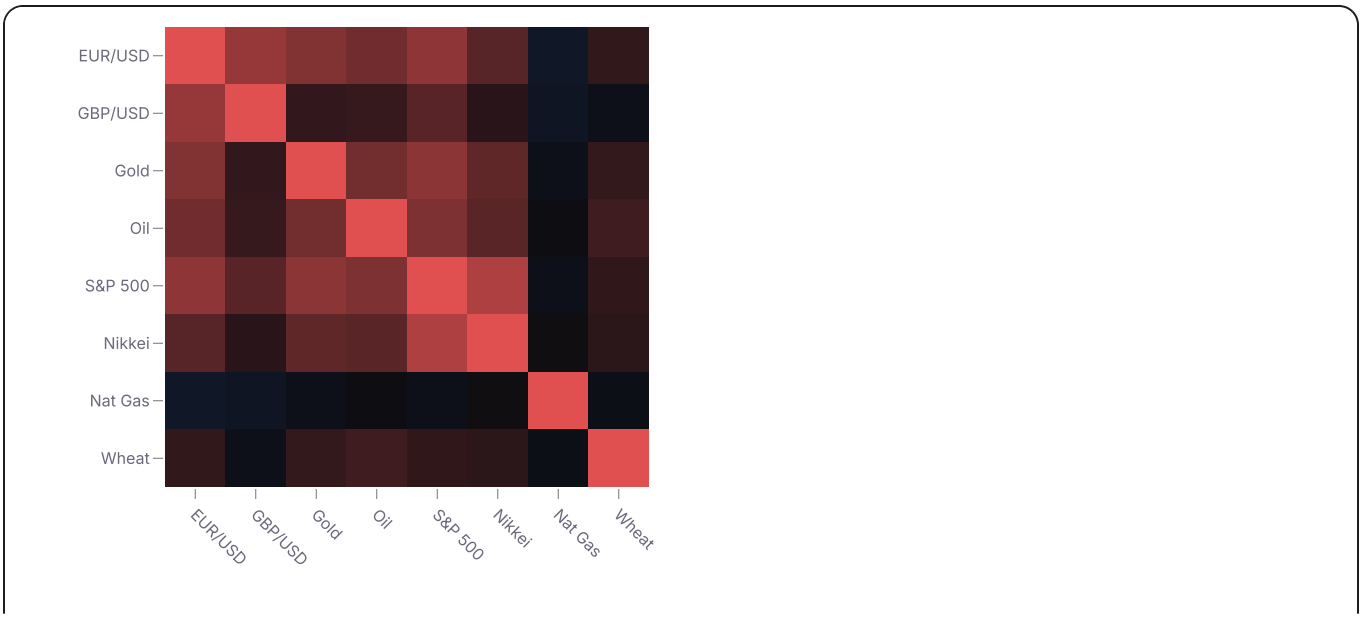



> ***What to look for:*** *Did all markets go red simultaneously, or did stress migrate from one to another? Oil's crisis (March 9, Saudi price war) hit before the broader equity crash. Gold — usually a safe haven — also saw liquidity deteriorate as traders sold everything for cash.*

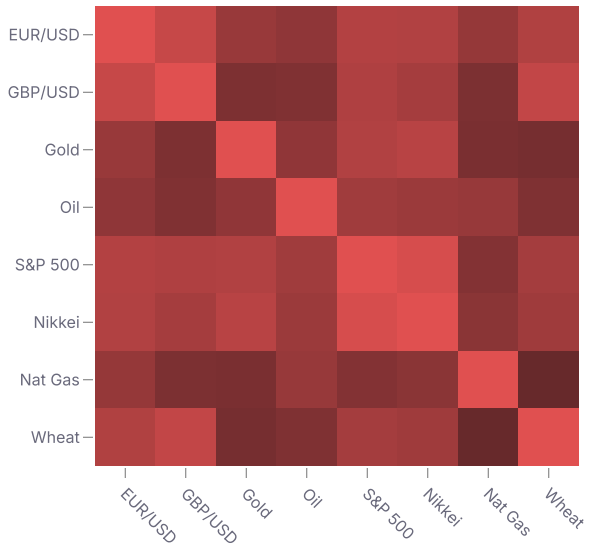
### D.2 — Correlation Shift: Calm vs Crisis

During normal times, markets have moderate, varied correlations — some move together, some don't. During a crisis, everything correlates. The three matrices below show: calm period, crisis period, and the difference.

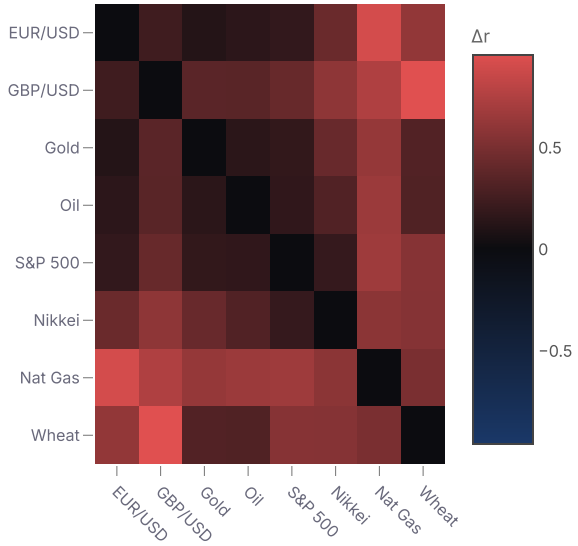
CALM (JUN–DEC 2019)



## CRISIS (FEB–APR 2020)




## DIFFERENCE (CRISIS – CALM)

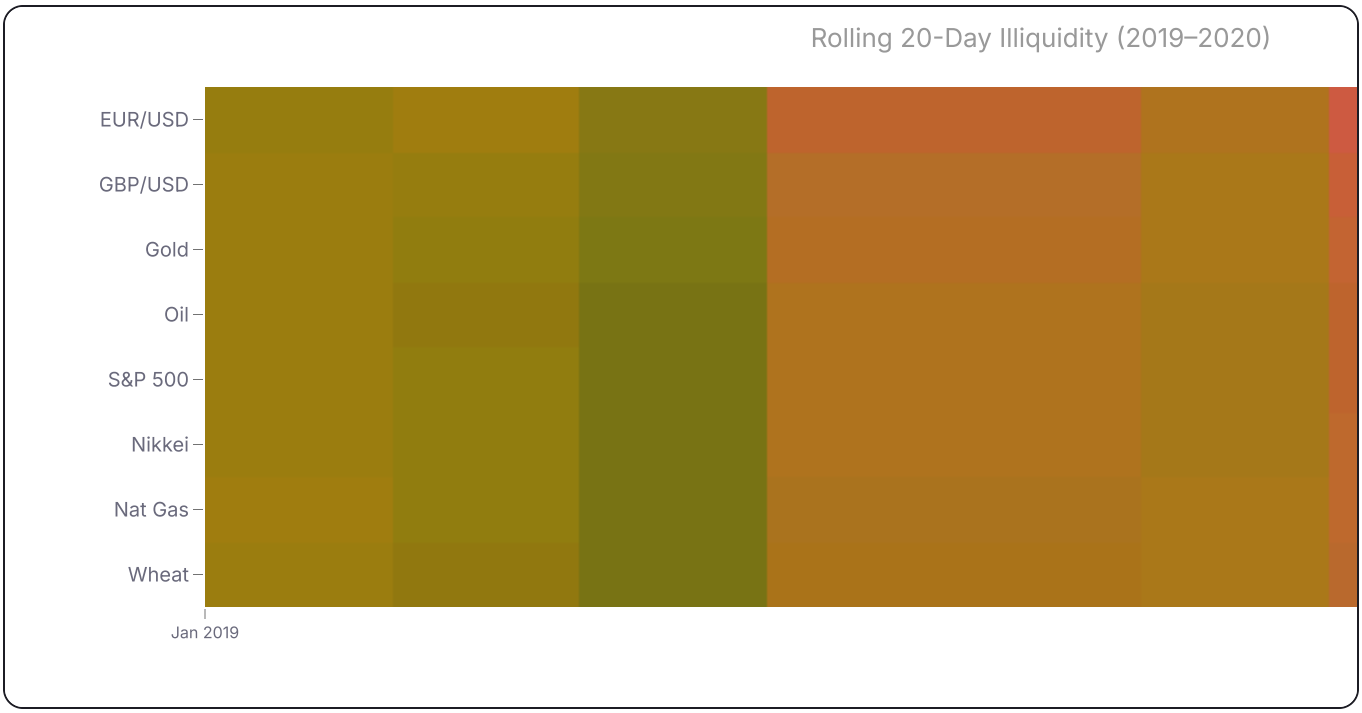


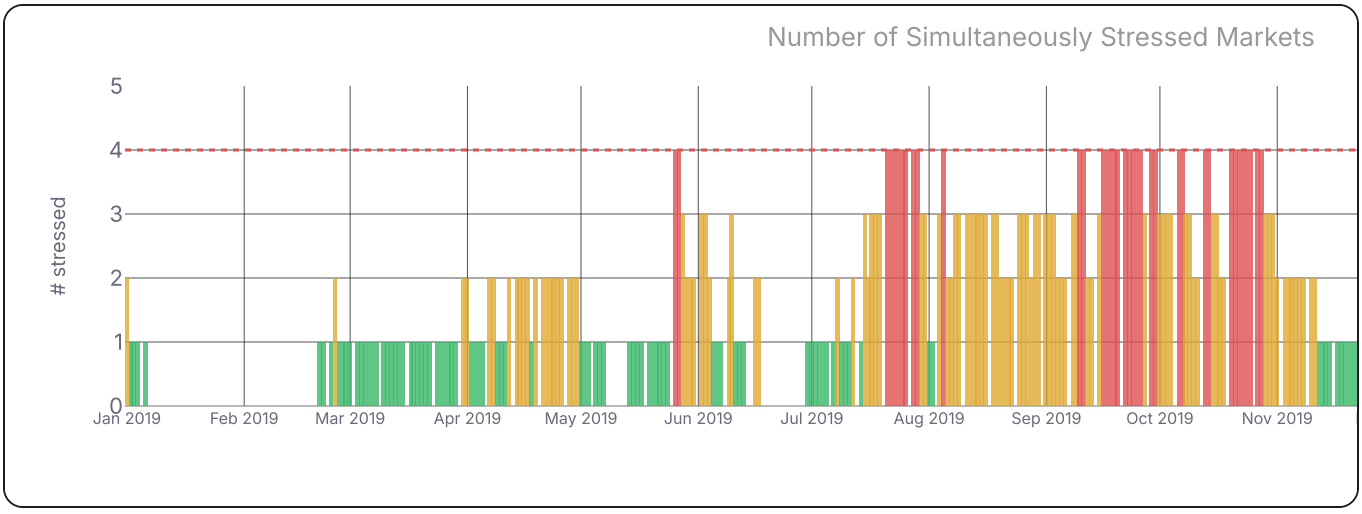

> ***The key insight:*** *In calm times, Gold and Oil might have low correlation with equities. During COVID, correlations jumped toward 1 across the board. This is "correlation breakdown" —*

> *diversification fails exactly when you need it most. The difference matrix makes this starkly visible.*

### D.3 — Systemic Stress Timeline (2019–2020)

The top heatmap shows each instrument's rolling illiquidity. The bar chart below counts how many markets are simultaneously stressed. When the bar crosses the red threshold (4+ markets), the system is in systemic stress.




> ***Systemic vs idiosyncratic:*** *Most illiquidity events are isolated — one market spikes while others are calm. But March 2020 should show a clear vertical stripe across ALL instruments*

> *simultaneously, with the stress count hitting maximum. That's systemic risk — and it's the scariest thing in finance.*

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InfoViz 247 | Market Liquidity